

Machine-Learning Driven Strategies for Adapting Immunotherapy in Metastatic NSCLC

Corresponding Author: Dr Jia Wu

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This is a nicely written paper that touches an important and practical clinical issue in cancer immunotherapy for NSCLC. As chemotherapy has inadvertently many side effects, it is important that some predictive variables or patient characteristics can be identified to guide the treatment decision. This paper presented a data driven and rigorous approach for this purpose. I have only a few minor comments:

1. As the authors mentioned, there are a number of data analytic approaches for discovering predictive variables and recommending treatments. Some rationale should be provided on why they chose to use the weighted approach.
2. I don't think I have actually seen the final prediction rules from the methods. It is nice to report them as well in addition to important variables
3. The overlap/non-overlap Venn diagram of the variables based on the three methods in Figure 7 is nice to see. However I am curious about how much overlap there is for the actual prediction rules. Of course, the rules are different but maybe they can be applied to both the training and the test data and see whether the same subjects were recommended to the corresponding treatments
4. The weighting and other methods also offer bias correction in absence of external testing data when evaluating results. I wonder if such correction can be applied to Table 2 which is based on the training data.

(Remarks on code availability)

The authors did not provide codes

Reviewer #2

(Remarks to the Author)

This manuscript uses methodology from the causal inference field to identify subgroups of NSCLC patients that are most likely to benefit from ICI monotherapy or ICI+Chemo combination therapy. The algorithms suggested by the authors fit statistical models that can provide "therapy recommendations" for future users. Upon fitting such models on a larger "discovery" cohort, they compare model performance on "validation" cohorts and claim that 3 of the 5 utilized loss functions would lead to meaningful improvements. They then move on to characterize the features utilized by these successful prediction models, which could be interesting for biomarker development.

I commend the authors for applying sound causal inference methodology in this field. These methods could be interesting for the development of personalized medicine. However, I have a few reservations with the work as it currently stands.

(1) The paper uses methodology that has been developed by others and is readily available for use in a convenient R package, "personalized"; the contribution of this paper is the application of said methodology to personalization of cancer treatments. There is nothing wrong with that, but the author occasionally use language that obscures this important fact. For instance, the statements "the study introduces an innovative approach" or "this methodological framework aims to revolutionize treatment selection" in the abstract can be easily misunderstood; and there are further examples like this

throughout the paper.

(2) The causal inference (CI) methodology is not fully explained. Although the authors do make significant efforts to introduce standard CI methodology such as propensity scores, the paper becomes more silent when it arguably starts to matter the most, i.e., when the specific features of this method are introduced. It gets especially confusing when it comes to the validation, because the main validation metric is the "treatment effect conditional on subgroups" which seems to be an average of subgroup-specific treatment effects, although I could not find the definition anywhere. Generally speaking the method only started making sense to me once I read references 10 and 11. Since I thought I was reasonably literate about CI, I wonder, then, how easy this will be to understand for the broader audience of Nature Communications.

(3) The CI methods are based on some important assumptions such as: no residual confounding, SUTVA, etc. These assumptions are not stated or discussed anywhere in the manuscript as far as I can tell, which would be important to do because they may not always be reasonable in practice.

(4) Related to this, the use of the word "validation" in this study is problematic. The methodology hinges on estimating individual/group-specific treatment effects (ITE), which is only possible under strong assumptions (see (3)). The "validation" performed in this paper consists of estimating ITEs in a different dataset and then showing, essentially and if I get this right, that ITEs are higher for people who actually received the recommended treatment. However this doesn't "validate" the model in the sense that the assumptions used to construct it are not checked. A more direct validation would need to use randomized data (could be retrospective), and would need to show that the actual treatment outcomes of patients who received the recommended treatments are better than those who did not. In other words, the authors would need to follow standard practice for establishing a biomarker (e.g. <https://ascopubs.org/doi/10.1200/JCO.2015.63.3651>). If the "Mayo cohort" is actually from a randomized trial, then such a direct validation would be possible.

In general, the paper should always emphasize that quantities like "difference of average treatment effect" are model-inferred quantities, rather than measured quantities.

(5) To the extent that validation was performed, there seems to be no check whether the "treatment effects conditional on subgroups" are statistically significant. It seems like the "personalized" package does provide methods to perform such tests. Additionally, correction for multiple testing could be needed given that 5 different loss functions were tried.

A few minor comments:

-- For propensity score matching, it would be more customary to show the imbalance of the `_covariates_` (and not the propensity scores themselves, as done in Figure 5) before and after matching.

-- There can be overlap between prognostic variables and treatment moderators, correct?

-- The reference to "three scoring benefits and five loss functions" in the abstract is not really understandable without having read the paper.

-- Is figure 1 really needed? Personalized treatment is not really a novel concept anymore these days

-- The authors choose to identify 3-months progression as a binary endpoint. I wonder why this choice was made rather than using the full survival data, since the R package "personalized" supports time-to-event outcomes.

-- Are the p-values reported in Table 1 for the original unmatched MDACC cohort? (E.g there is no apparent difference between Line of ICI but the p-value of 0.002 is very low)

-- The choice of loss function quite dramatically impacts how many people are selected for each treatment. For instance, with the linear model, the majority are recommended to receive ICI-Chemo.

(Remarks on code availability)

The authors stated that they will only make the code available upon publication (p. 25) so I was unable to review the code.

Reviewer #3

(Remarks to the Author)

The overarching goal of this paper is to improve treatment selection for individual patients using baseline covariates. Specifically, the clinical objectives of this study are to assess the impact of adding chemotherapy on early progression outcomes in NSCLC patients receiving ICI therapy. The paper is well written and the authors utilize many elements of modern statistical practice including the use of propensity score matching; the use of a discovery sample for model development and an external-validating testing sample for assessing how well the models do with out-of-sample predictions, i.e. subgroup identification; and the use of sensitivity analysis, including several different model specifications for loss functions.

An issue for this reader with respect to the field of personalized medicine in general is the use of terminology, such as "personalized immunotherapy" or the expression of an objective to seek "optimal personalized treatment strategies" implying

that the outcome of this work will be the identification of the optimal treatment for an individual. In fact, as the authors rightly note (p.3: L 88) "The process of determining the best treatment allocation can be viewed as a task of subgroup identification, wherein subgroups are identified based on the heterogeneity in the treatment effects." What is achievable statistically is to identify a sub-group of patients who share common characteristics and investigate what treatment works best for such subgroups. In other words the "optimal personalized treatment strategy" is the treatment for the canonical individual who best represents a sub-group.

The technological sophistication of this work is quite high consisting of multiple components. One element of this program is the specification of different loss functions. The authors highlight that (p. 11: L248) "[T]he analysis shows the influence of various loss functions on predictive biomarker identification for early progression in patient cohorts undergoing ICI-Mono or ICI-Chemo." See figure 6. They write (p.12: L275):

"This investigation yielded a striking revelation: There are a rather limited number of overlapping features among the diverse loss functions, strongly suggesting that the distinct loss functions are effectively capturing discrete information or intricate patterns present within the data. This inherent diversity in the loss functions' capabilities resulted in the identification of distinct sets of biomarkers, each carrying its own valuable insights."

One wonders whether this is in fact the only explanation and whether this result is what the authors expected. Would the use of other loss functions have identified other sets of biomarkers that would also carry valuable insights? From the point-of-view of sensitivity analysis wouldn't one have hoped that the results were robust to the specification of different loss functions? In the end they do report a singular predictive biomarker – the mutated variant EGFR, that impacts treatment response for patients treated with ICI-Chemo, whereas among patients undergoing ICI-mono treatment, the use of tobacco emerged as a predictive factor. One wonders if these predictive factors could have been identified using less sophisticated methodologies? In other words, are there alternative approaches for identifying patients prone to early progression within treatment groups and the biomarkers that are associated with treatment response?

(Remarks on code availability)

Reviewer #4

(Remarks to the Author)

The authors have devised an approach to selecting patients for optimal treatment choice. This leverages various models to determine the 'individualized treatment rules' for each patient, to find whether they would derive most benefit from ICI (an acronym they do not define in the abstract) or chemo/ICI. They leverage a 750 patient cohort and validate their approach in a further 80 patients.

Unfortunately, it is of the Reviewer's opinion that this manuscript should be rejected.

The observation that smokers appear to benefit more from ICI-mono than ICI-chemo is moderately interesting. They do not validate this observation, instead focussing on validating their approach for patient selection.

As their Figure 1 demonstrates, for precision medicine to be effective, there must be multiple theoretical options available that, if given, would provide a patient with meaningful clinical benefit. This is not the case in this (or any) indication. They present this problem as the focus of 'a substantial effort with the statistics and machine learning community' when really, it is a basic science / drug discovery / drug development / trials / data availability problem (even without acknowledging the problem of pre-existing or acquired resistance).

Therefore, the novelty here must come from the uniqueness of the modelling framework they propose, or the data they present. Regrettably, the data is of limited utility, as they are not proposing sharing it (available from the corresponding author 'upon request'), they don't define what it is in the Methods (was it WGS/WES/a panel?), and the modelling framework is very opaque, poorly explained and with no code available to review, and of dubious/no benefit.

Further points:

Their study hinges on their ability to identify and control for bias arising from intrinsic differences between patients selected for ICI-mono and ICI-chemo – the reviewer is not convinced that with their limited clinical data, that does not for example include Performance Status, they would be able to do this meaningfully.

There is no description of how the genomic mutation markers were derived, or what they were, anywhere in the paper.

This paper does not consider the possibility, which would be supported by the related literature, that they are testing the wrong covariates and that none of the markers they are testing are likely to be biomarkers for treatment as they propose (I think this is likely – I would focus on ICI-related biomarkers identified in Litchfield et al's meta analysis from Cell 2021).

The fact that their various loss models find such divergent results from the same data makes me question their value.

There is no indication of what would constitute 'success' as far as their validation cohort is concerned. I could not ascertain a

p value or understand how much improvement one might expect by chance. The opaqueness of their methods make me the more convinced that this approach would require validation in a prospective clinical setting.

The figures are not nearly of a standard that would be expected for publication in this journal.

(Remarks on code availability)

There was no code provided.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have addressed many of my comments. The additional external validation data is a nice plus. While I appreciate that the authors cited many subgroup identification methods in the introduction section, I found quite a few of them are only remotely related. It might be a distraction and can actually invite more questions. For example, the references 20-27 are more about biomarker research. Although related, this is not the focus of this paper.

(Remarks on code availability)

I reviewed the codes very quickly and did not have the opportunity to test the codes

Reviewer #2

(Remarks to the Author)

In this revision, the authors have substantially extended their prior work. Notably, two further large cohorts were added to this study, Dana-Farber and SU2C, and three new prediction methods were also added. The explanation of the methodology has also been improved a lot compared to the previous version. Overall, the authors invested a significant amount of work in this revision, aiming to address as many comments as they could. Unfortunately, another major issue has now been introduced in this revised version, and another major issue has not really been solved.

-- All five methods that were proposed in the original version of this manuscript failed to work in the two validation cohorts. I am afraid that this was the reason why the three other new methods (H-STEP, S-STEP, and A-STEP) were also added, although that is not made clear. Of these three new methods, only A-STEP can be "validated" in the two new cohorts; however, the p-values for the two arms are 0.016 and 0.0076. Since the authors have now tried 8 methods in total on this data, some kind of p-value correction should be applied to correct for this fact; in that case, these p-values may not remain significant.

So, in total, it could be questionable that the new proposed A-STAR method -- which seems to have been developed using the validation data more than once -- would actually generalize.

-- The methods are tested by determining the survival in the cohorts of those patients who receive the treatment recommended by the model to those who were not treated as recommended by the model. As remarked in the previous manuscript, direct validation of this kind of prediction would require analysis of a _randomized_ rather than retrospective / observational trial. While the authors do employ propensity score matching in an attempt to reduce confounding, we still need to buy that there is no residual confounding, which can be questionable since some important clinical variables seem to be missing (see comment by R4). Unfortunately, this point was not really addressed by the revision. There now is some limited discussion of this issue, but still the claims are presented in stronger language than the results really allow (e.g. a reader would be forgiven to assume that a "proof-of-concept study" of "personalization of immunotherapy" would involve some actual personalization).

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The authors have significantly revised their original submission based on the feedback from the referees. As a result this revision is much improved and in effect is a brand new paper. I have 3 comments.

(1) Please remove any reference to "artificial intelligence" in the title, abstract and in the paper. You are using machine learning which you note repeatedly throughout the paper. The need to refer to AI is unnecessary. I would strongly encourage you to delete the reference.

(2) The methods being used are quite sophisticated and details are provided in the supplementary material. I leave it to the Editors to decide if this is an appropriate article for the journal.

(3) Referee 4 made an important observations regarding the original tone and scope of the work In response the authors wrote:

"Our study is not intended to solve the basic science/ drug development challenges of precision treatment directly, as you correctly pointed out. Our research serves as a proof-of-principle demonstration of how machine learning approaches can be leveraged to address these complex clinical decisions as they arise, particularly in contexts where treatment efficacy cannot be linked to a simple e.g. genomic biomarker . . . Specifically, we used the ICI-Mono vs ICI-Chemo question as an illustrative example to showcase the potential utility of our approach in guiding treatment decisions, particularly in situations where clinicians face a lack of head-to-head trial data.

We understand and agree that our initial manuscript may have overstated the immediate clinical applicability of our findings. To address this, we have revised the manuscript to clarify that our aim is to complement ongoing drug discovery efforts by providing a statistical and machine learning framework that can help prioritize and evaluate potential treatment options based on patient-specific data. We recognize that our work is one piece of a larger puzzle and does not negate the critical role of basic science, drug development, and clinical trials."

I agree with these comments completely and would like the authors to convey this sentiment at the outset of the paper in the abstract and in the Introduction.

(Remarks on code availability)

Version 2:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

The authors have, in response to my and other reviewers' earlier comments, again softened the language used in the paper, which now avoids strong terms such as "treatment personalization". I appreciate that these changes were made.

The most important issue I had raised in the previous iteration was the fact that 3 new methods were introduced in that version, as the 5 original methods presented in the initial version failed to work on the validation dataset. While the authors seem to agree with this assessment, they respond that the three new methods should in fact not be considered "new methods" since they are ensembles of the earlier ones. I disagree with this statement because in ensemble learning one introduces additional parameters that can be fitted, meaning that the capacity of the models increases and they can generate different results. The new version of the paper now presents this approach as if it had been the plan all along, and does not clearly state that these models were only designed after the initial ones failed to work (they were not in the original version of the manuscript).

In addition, the authors have performed a p-value correction. I am unable to verify this calculation as I cannot find the individual p-values reported anymore, nor information on which models were included in this correction (just the final 3 or all 8?). However clearly the corrected p-values show that the fact that only 1 out of the 3 ensemble models reaches significance (in fact only 1 out of the 8 total tested ones) does not remain significant once the amount of tested models is taken into account. The authors choose a relatively lenient correction method (FDR instead of Bonferroni) and an unusual "significance" (in fact FDR) threshold of 0.1 to state significance anyway.

In total, the current manuscript version still presents a somewhat overly positive view of the authors' findings, and the evidence presented that this approach actually works is not particularly strong from a statistical perspective.

Version 3:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors are very responsive and have addressed the raised concerns by putting in clear and cautionary languages in the main text.

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Response to the reviewers

We thank the reviewers for their time to review our manuscript and provide excellent suggestions to further improve it. Please kindly note that to comply with all reviewers' requests, we have now restructured both study participants (with the addition of 1,470 samples from two additional sources) and experiment design. We have now included samples from three cancer centers: MD Anderson Cancer Center (n=750), Mayo Clinic (n=80), and Dana-Farber Cancer Institute (n=1077) along with one multi-site consortium from Stand Up To Cancer (n=393) totaling 2300 samples. In terms of experiment design, three main changes have been introduced: (1) an ensemble learning scheme was used to combining different weighting methods, (2) repeated cross validation was used to optimize weighting algorithms, which were further tested on external dataset, and (3) beyond predicting early disease progression status at 3 months, the model's predictive performance was assessed based on progression free survival data. These have resulted in 4 new figures (Figure 1-4), 2 new tables (Table 2-3), 15 new supplementary figures (Suppl Fig S1-S15) and 1 new supplementary table (Suppl Table S1). We believe these changes have significantly strengthened the manuscript and hope you will now find it suitable for publication.

Below are our responses to the reviewers' comments. We are more than happy to make any further revisions recommended by the reviewers.

Reviewer #1

This is a nicely written paper that touches an important and practical clinical issue in cancer immunotherapy for NSCLC. As chemotherapy has inadvertently many side effects, it is important that some predictive variables or patient characteristics can be identified to guide the treatment decision. This paper presented a data driven and rigorous approach for this purpose. I have only a few minor comments:

1. As the authors mentioned, there are a number of data analytic approaches for discovering predictive variables and recommending treatments. Some rationale should be provided on why they chose to use the weighted approach.

Thank you for your insightful comments. The idea behind the use of the weighted approach (to be exact, the propensity score weighted loss function) is similar to how inverse probability weighting (IPW) works, wherein weights are applied to each observation or individual in the analysis. In our case, it simply means the weight (the propensity score) is utilized as an additional variable to be adjusted when estimating benefit scores of adding chemotherapy for each patient via the constructed loss function. The key reason is to remove any potential bias in the results of real-world data-based studies by giving relative 'importance' or 'weights' to each observation (i.e. greater weights to under-representative groups), aiming to enhance the generalizability of the conclusions drawn from the studied population onto the broader population.

There are indeed other methods available to tackle this challenge, for example covariate adjustment through a regression modeling, where all relevant patient characteristics are included in regression modeling. However, a commonly cited concern is that such techniques tend to overfit when the number of covariates is large while the studied population is small. We have now included this rationale under **Supplementary Methods S3: The Weighting Method**.

"A significant advantage of the weighting method is that, even when complete outcome regression models are employed, the validity of the resulting estimators remains

unaffected by potential misspecifications in the said models. On top of this, the weights represent the relative ‘importance’ of an observation which is useful in balancing a non-randomized cohort, subsequently removing any potential bias.”

- I don't think I have actually seen the final prediction rules from the methods. It is nice to report them as well in addition to important variables.

Thank you for your suggestion. We have now introduced an ensemble model called A-STEP by aggregating five scoring model's predictions previously using the attention-based mechanism. **New Fig. 1** shows the overall design (the red highlighted box is the ensemble part) and **Updated Fig. 5** (next page) shows the proposed model's interpretation in terms of the association of individual features and the model stratification. It is challenging to visualize the final prediction rules. Beyond the feature importance, we also presented the details of individual models as well as their integration in **Suppl Methods S3 & S4**. In addition, we shared the model codes and datasets to facilitate interpretation and reproducibility.

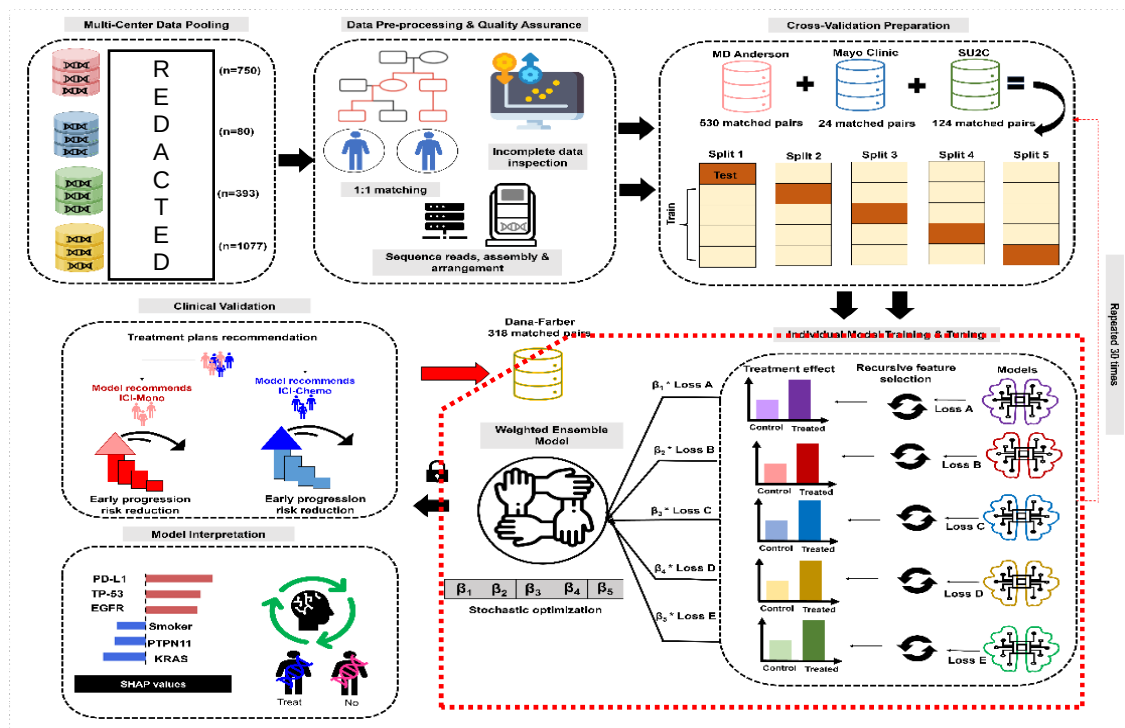


Figure 1 A-STEP diagram.

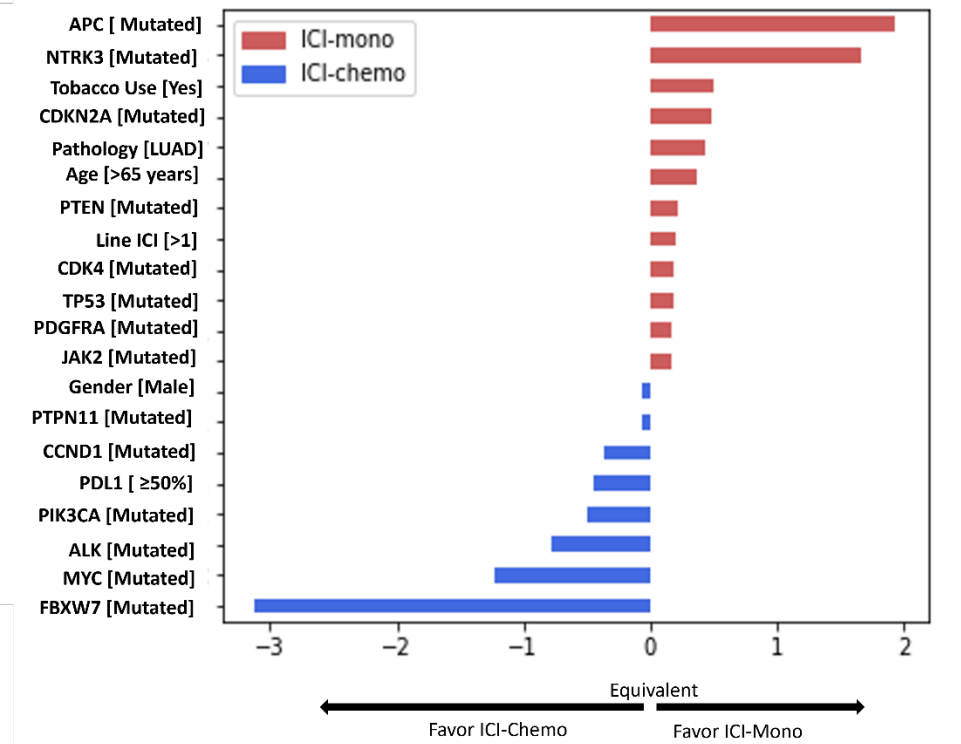
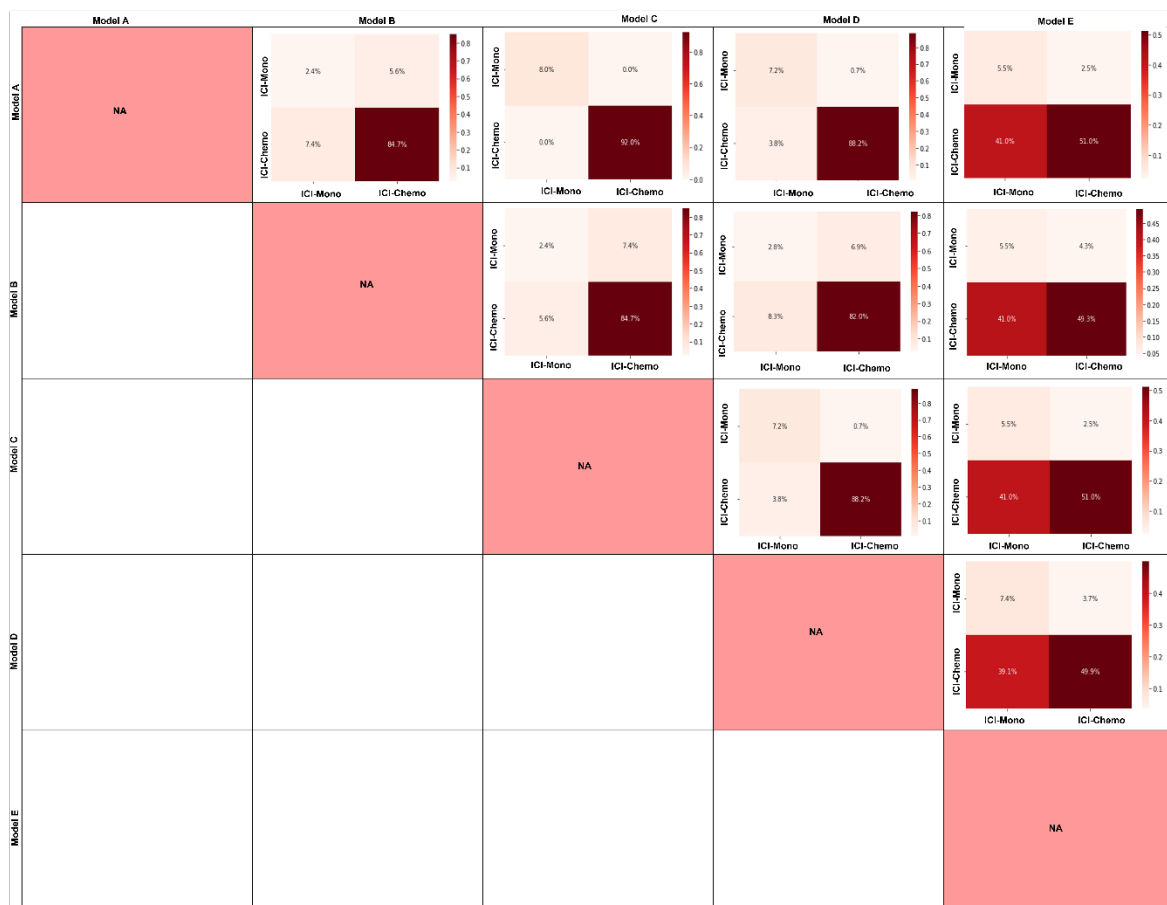


Figure 5: A-STEP SHAP interpretation.

3. The overlap/non-overlap Venn diagram of the variables based on the three methods in Figure 7 is nice to see. However, I am curious about how much overlap there is for the actual prediction rules. Of course, the rules are different but maybe they can be applied to both the training and the test data and see whether the same subjects were recommended to the corresponding treatments.

Thank you for your thoughtful suggestion. We have now incorporated a pair-wise correlation matrix showing the overlap of treatment recommendation between different standalone models. The below figure is now added as the new **Supplementary Figure S13**.



Supplementary Fig S13 Treatment recommendations overlapped between sub-models

4.The weighting and other methods also offer bias correction in absence of external testing data when evaluating results. I wonder if such correction can be applied to Table 2 which is based on the training data.

Thank you for your insightful comment regarding the application of bias correction methods. We have indeed considered bias correction during model training. In addition, additional samples from Dana Farber Cancer Institute (DFCI) were collected as the external validation cohort to ensure the robustness and generalizability of our predictive models. The inclusion of these new datasets allows us to better evaluate the performance of our models and mitigate overfitting risk. Taken together, these enhance the credibility and rigor of our study. Thank you once again for your valuable suggestion, which has significantly contributed to the improvement of our manuscript.

5.Reviewer #1 (Remarks on code availability): The authors did not provide codes.

Thank you for your remark regarding the availability of codes. We appreciate the importance of transparency and reproducibility in research. In response to this valuable feedback, we have made our codes publicly available. The codes used for data analysis, model development, and evaluation are now accessible through a dedicated repository on [<https://github.com/WuLabMDA/A-STEP>]. This repository includes detailed documentation and instructions for replicating our study's results. We believe that sharing our codes will facilitate further research and collaboration in the field, and we are committed to supporting the scientific

community by providing the necessary resources for verifying and extending our work. Thank you for your suggestion, which has helped us improve the openness and reproducibility of our research.

Reviewer #2

This manuscript uses methodology from the causal inference field to identify subgroups of NSCLC patients that are most likely to benefit from ICI monotherapy or ICI+Chemo combination therapy. The algorithms suggested by the authors fit statistical models that can provide "therapy recommendations" for future users. Upon fitting such models on a larger "discovery" cohort, they compare model performance on "validation" cohorts and claim that 3 of the 5 utilized loss functions would lead to meaningful improvements. They then move on to characterize the features utilized by these successful prediction models, which could be interesting for biomarker development.

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1. The paper uses methodology that has been developed by others and is readily available for use in a convenient R package, "personalized"; the contribution of this paper is the application of said methodology to personalization of cancer treatments. There is nothing wrong with that, but the author occasionally use language that obscures this important fact. For instance, the statements "the study introduces an innovative approach" or "this methodological framework aims to revolutionize treatment selection" in the abstract can be easily misunderstood; and there are further examples like this throughout the paper.

Thank you for your valuable feedback. We appreciate your recognition of the methodology we employed and your emphasis on the importance of clear and accurate language in our manuscript. Acknowledging this important comment, we have revised the language. In addition, we changed the study design by proposing an ensemble method called A-STEP, which innovatively fuses different weighting algorithms through grey-wolf optimization-based attention mechanism to address the limitations of individual models by combining their predictions, leading to a more accurate, robust, and generalizable model. The new **Fig. 1** below shows the pipeline of the overall study design. By making these changes, we aim to clearly communicate our contributions. Thank you again for your insightful comments, which have helped us to improve the clarity and precision of our manuscript.

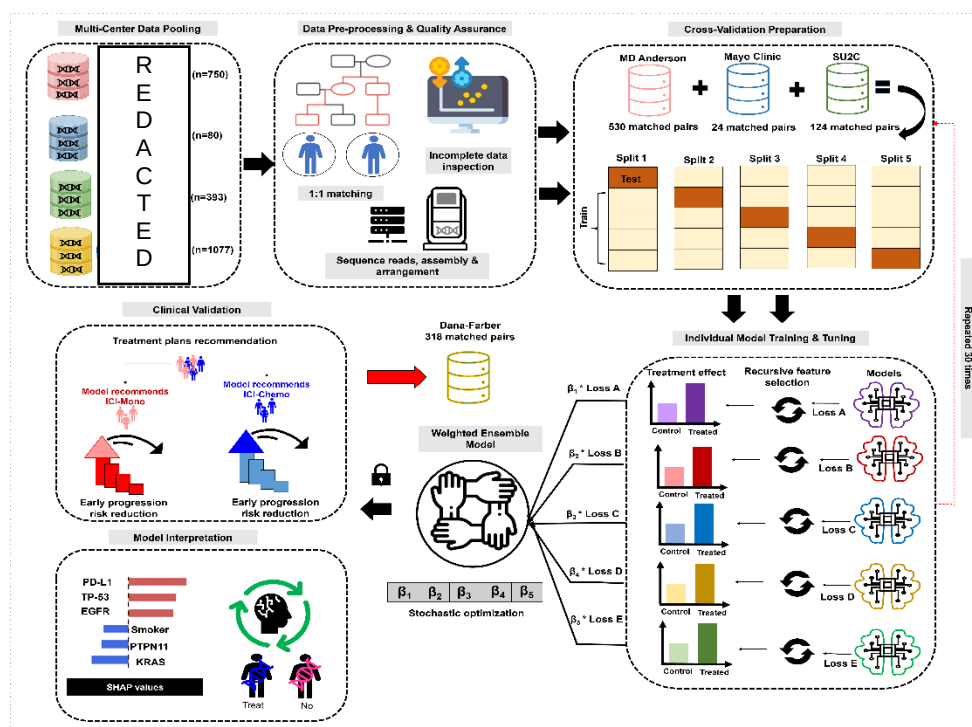


Figure 1: A-STEP diagram

- The causal inference (CI) methodology is not fully explained. Although the authors do make significant efforts to introduce standard CI methodology such as propensity scores, the paper becomes more silent when it arguably starts to matter the most, i.e., when the specific features of this method are introduced. It gets especially confusing when it comes to the validation, because the main validation metric is the "treatment effect conditional on subgroups" which seems to be an average of subgroup-specific treatment effects, although I could not find the definition anywhere. Generally speaking the method only started making sense to me once I read references 10 and 11. Since I thought I was reasonably literate about CI, I wonder, then, how easy this will be to understand for the broader audience of Nature Communications.

Thank you for your detailed and insightful feedback regarding the causal inference (CI) methodology presented in our manuscript. Following this comment, we have restructured our entire **Methods** section into several sections to better fit the *Nature Communications* audience, which would allow them to follow each step and reproduce the work: (1) how the model was developed using causal inference methodology, which has been divided into several key steps, (2) from the clinical translation perspective in which we have explained in details regarding the performance metrics used to represent the "treatment effect conditional on subgroups" in this work, (3) model interpretation and reasoning, and (4) how the models were trained and tuned. Appended are examples from the main draft. Please refer to the revised **Methods** section for the full details.

"Step 1: Modeling treatment effect"

In this work, we made standard assumptions shown to be sufficient for causal dependencies on the interventions (details in Suppl Methods S2). Without loss of generality and according to Rubin's potential outcome framework³⁸, the expected

outcome E can be expressed given the desired outcome (O), treatment (T), and initial covariates C as $E[O|T, C] = \phi(C) + T \times \Delta(C)/2$, where $\phi(C)$ is a function that represents the main prognostic effect and can be calculated as $E[O|T = 1, C] + E[O|T = 0, C]$ and $\Delta(C)$ indicates the treatment effect which can be expressed as $E[O|T = 1, C] - E[O|T = 0, C]$."

"Step 2: Individual benefit score estimation through weighting method"

The objective was to utilize the weighting subgroup identification method^{10,11} to develop a personalized benefit scoring system $f(C)$ based on the covariates C , in which personalized treatment would be suggested for individual patients using $f(C)$. To clarify, we defined a benefit score as any function $f(C)$ that met two criteria: (i) it reflects the extent to which patients benefit from a certain treatment and (ii) it has a meaningful and known cut-point value c , such that for a given level of covariates C , $f(C) < c$ indicates that ICI-Chemo is more effective than ICI-Mono, and $f(C) \geq c$ indicates the opposite."

3. The CI methods are based on some important assumptions such as: no residual confounding, SUTVA, etc. These assumptions are not stated or discussed anywhere in the manuscript as far as I can tell, which would be important to do because they may not always be reasonable in practice.

Thank you for your insightful feedback. We have now revised the manuscript as suggested by explicitly stating and discussing these assumptions which can be found in **Supplementary Methods S2**. Below is the screenshot for your reference.

"S2: The Causal Inference Assumptions"

Our causal inference methods rely on several important assumptions which are crucial for the validity of our analysis. These assumptions are:

- *No Residual Confounding: We assume that all relevant confounders have been measured and included in our model. This assumption is critical to ensure that the treatment assignment is independent of potential outcomes given the covariates. In practice, this may not always hold true, and any unmeasured confounding can bias our results.*
- *Stable Unit Treatment Value Assumption (SUTVA): This assumption implies that the treatment of one patient does not affect the potential outcomes of another patient. This is important to ensure that the potential outcomes are only influenced by the treatment received and not by external factors.*
- *Treatment Ignorability: We assume that the treatment assignment is independent of the potential outcomes given the covariates. This means that the decision to assign a treatment is as if it were random given the observed covariates, allowing us to estimate causal effects accurately."*

4. Related to this, the use of the word "validation" in this study is problematic. The methodology hinges on estimating individual/group-specific treatment effects (ITE), which is only possible under strong assumptions (see (3)). The "validation" performed in this paper consists of estimating ITEs in a different dataset and then showing, essentially and if I get this right, that ITEs are higher for people who actually received the recommended treatment. However this doesn't "validate" the model in the sense that the assumptions used to construct it are not checked. A more direct validation would need to use randomized data (could be retrospective),

and would need to show that the actual treatment outcomes of patients who received the recommended treatments are better than those who did not. In other words, the authors would need to follow standard practice for establishing a biomarker (e.g. <https://ascopubs.org/doi/10.1200/JCO.2015.63.3651>). If the "Mayo cohort" is actually from a randomized trial, then such a direct validation would be possible.

We thank the reviewer for this comment. We have added two external datasets in this study, including Dana-Farber Cancer Institute (n=1077) and one multi-site consortium from Stand Up To Cancer (n=393). Following the reviewer's suggestions, the Dana Farber dataset was propensity score matched and used as our **external** validation cohort. To show that the actual treatment outcomes of patients who received recommended treatments were better than those who did not in the external validation cohort, we have now included two performance metrics: (1) the risk of early progression reduction as reported by the new **Table 3**, and (2) Kaplan-Meier survival curves with hazard ratios and p-values as reported in the new **Fig. 3**.

For the risk reduction shown in Table 3, we showed that A-STEP (the integrated model) showed a -16.0% reduction when patients were recommended to receive ICI-Mono and were indeed treated with ICI-Mono. Similarly, a reduction of -19.3% was observed when patients were recommended to receive ICI-Chemo and were indeed treated with ICI-Chemo.

Table 3: Refitted treatment effects reported as weighted risk reduction (WRR) in all sub-scoring and integrated models. Negative sign represented reductions in risk.

Model	Discovery		External Validation	
	ICI-Mono (%)	ICI-Chemo (%)	ICI-Mono (%)	ICI-Chemo (%)
A	-26.5 (n=54)	-15.1 (n=624)	13.6 (n=29)	-9.2 (n=289)
B	-3.9 (n=66)	-13.8 (n=612)	-9.2 (n=31)	-12.1 (n=287)
C	-26.5 (n=54)	-15.1 (n=624)	13.6 (n=29)	-9.2 (n=289)
D	-21.5 (n=75)	-16.2 (n=603)	21.4 (n=34)	-18.4 (n=284)
E	1.1 (n=315)	-20.1 (n=363)	-5.1 (n=115)	-18.8 (n=203)
H-STEP	4.0 (n=366)	-21.5 (n=312)	0.7 (n=142)	-18.4 (n=176)
S-STEP	-29.6 (n=56)	-15.7 (n=622)	2.2 (n=32)	-9.9 (n=286)
A-STEP	-13.5 (n=208)	-23.4 (n=470)	-16.0 (n=83)	-19.3 (n=235)

Additionally, we also showed that A-STEP stratification for progression free survival reinforced our findings as shown in Fig. 3 below, namely that patients had better outcomes when they received treatment as recommended by the model. This finding also aligned with standard practice as highlighted in <https://ascopubs.org/doi/10.1200/JCO.2015.63.3651>.

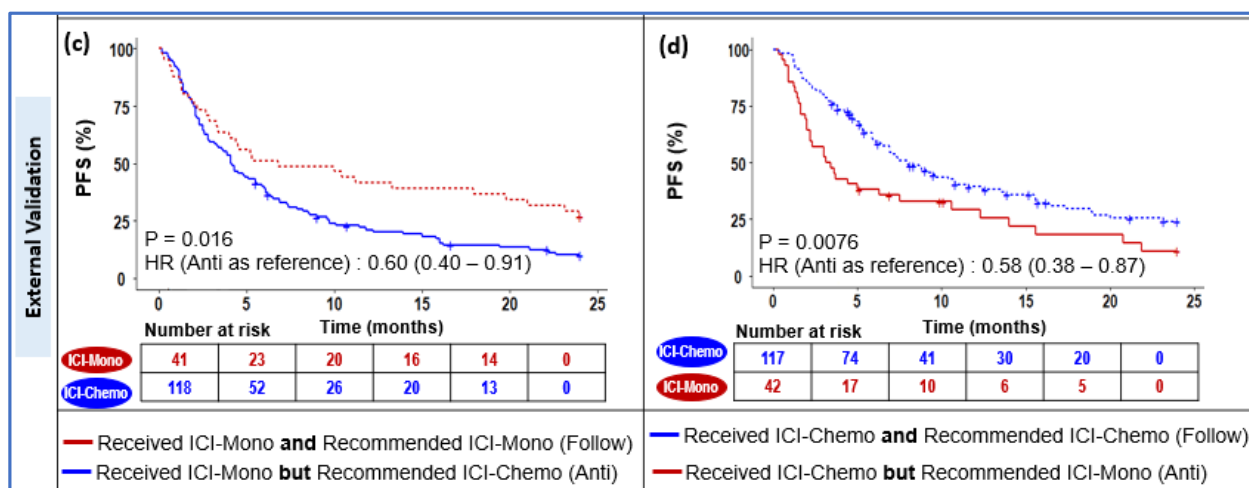


Figure 3: Comparison of 2-year PFS stratified by actual treatment plans.

5. To the extent that validation was performed, there seems to be no check whether the "treatment effects conditional on subgroups" are statistically significant. It seems like the "personalized" package does provide methods to perform such tests. Additionally, correction for multiple testing could be needed given that 5 different loss functions were tried.

Thank you for your constructive advice. As suggested, we have now included interaction plots with p-interaction to quantify the strength of treatment effects between stratified subgroups. For the integrated model, the new **Fig. 4** depicted the interaction plots of our ensembled model A-STEP for discovery and validation cohorts. By contrast, we also presented the interaction plot for 5 individual weighting models in **Supplementary Fig S8-S9**. We observed that A-STEP had substantial treatment effect present in discovery and external validation cohort as shown by the statistically significant interaction p values. However, the same observation did not hold in the stand-alone models.

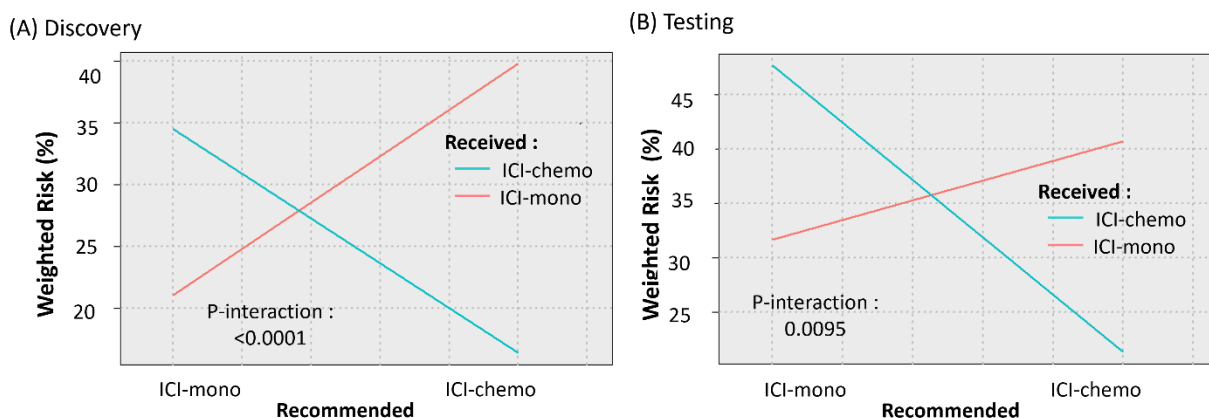
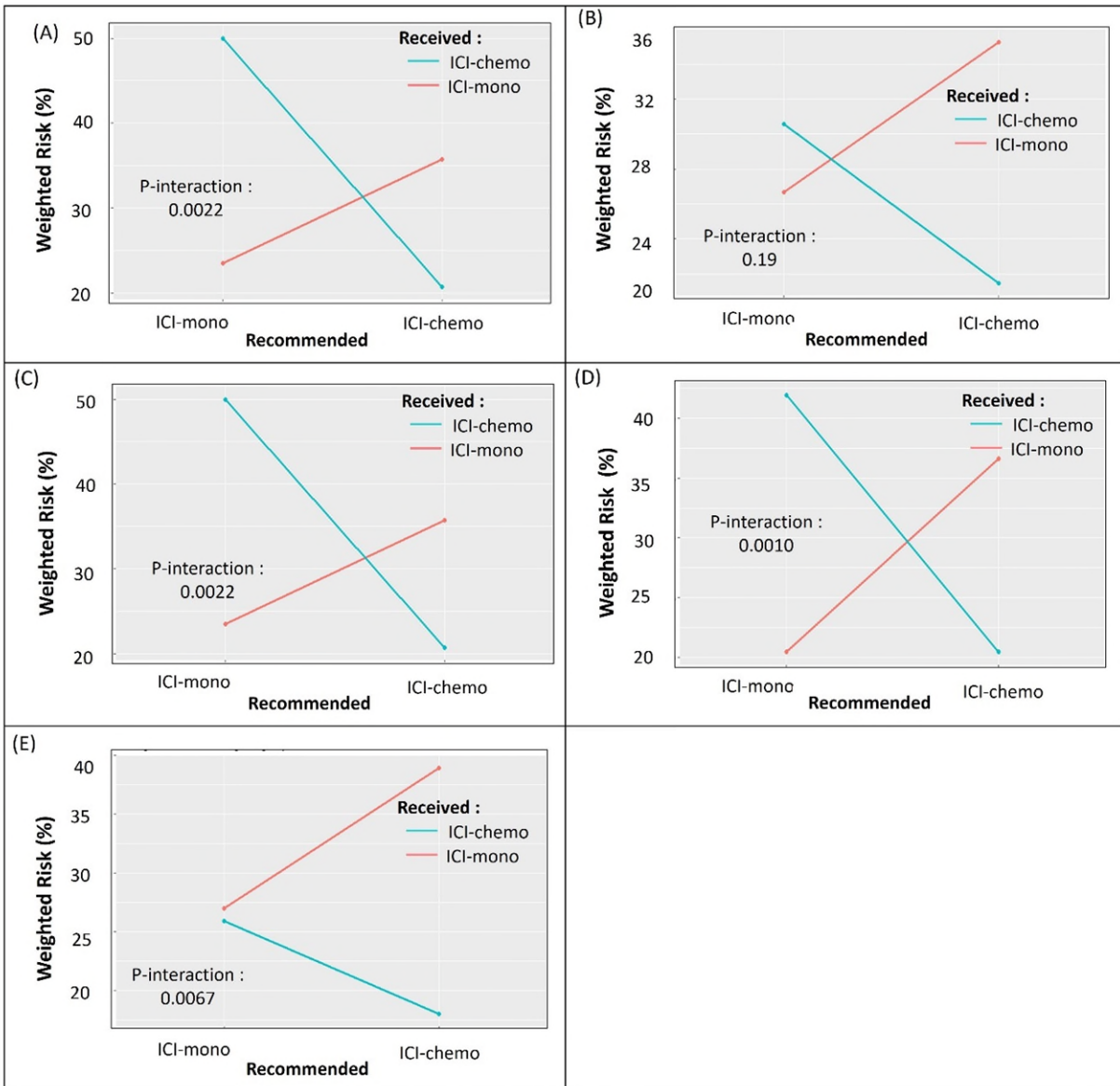
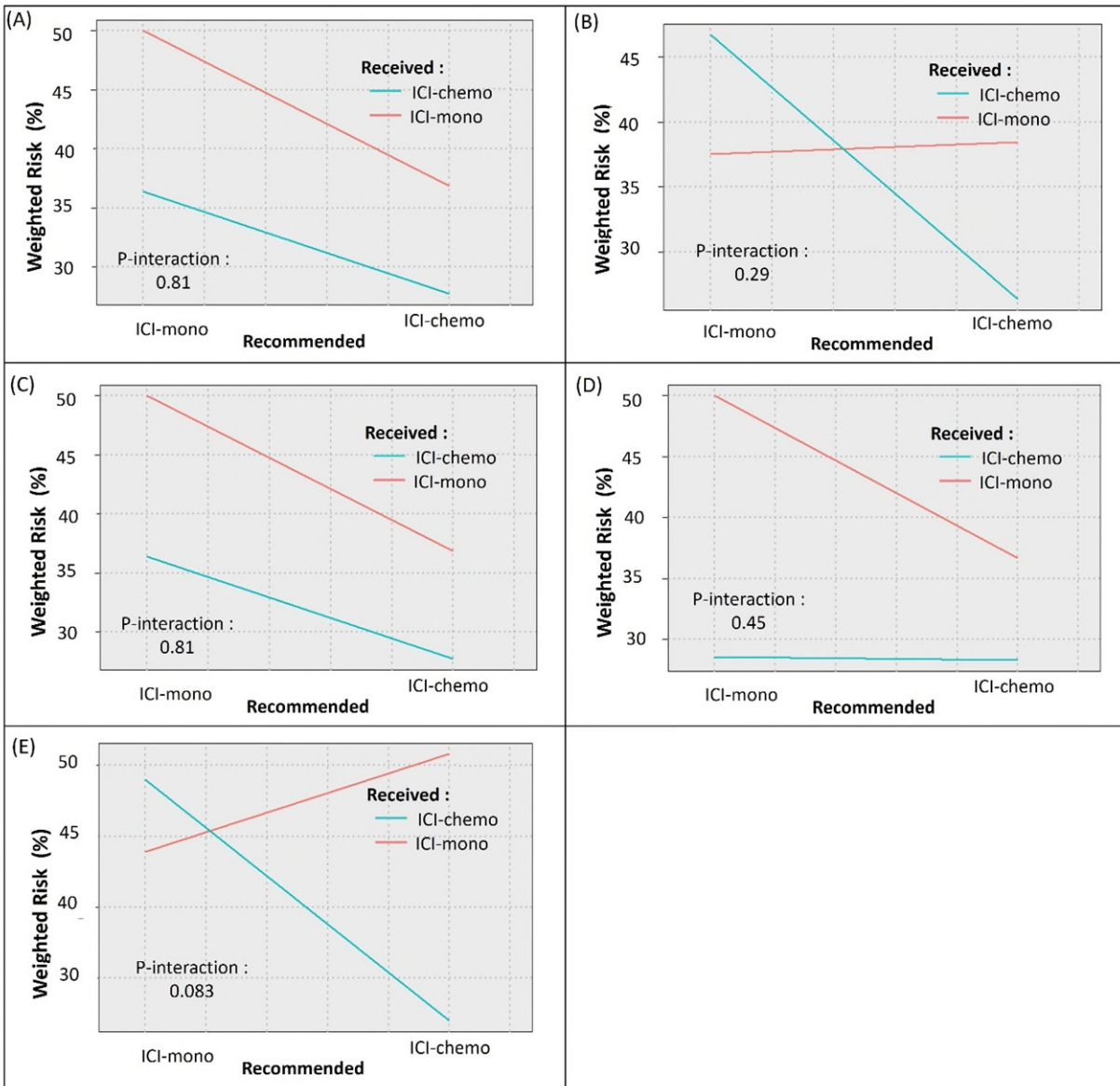


Figure 4: Interaction plots for (a) discovery and (b) external validation cohorts.



Supplementary Fig S8: Interaction plots for sub-models in discovery cohort

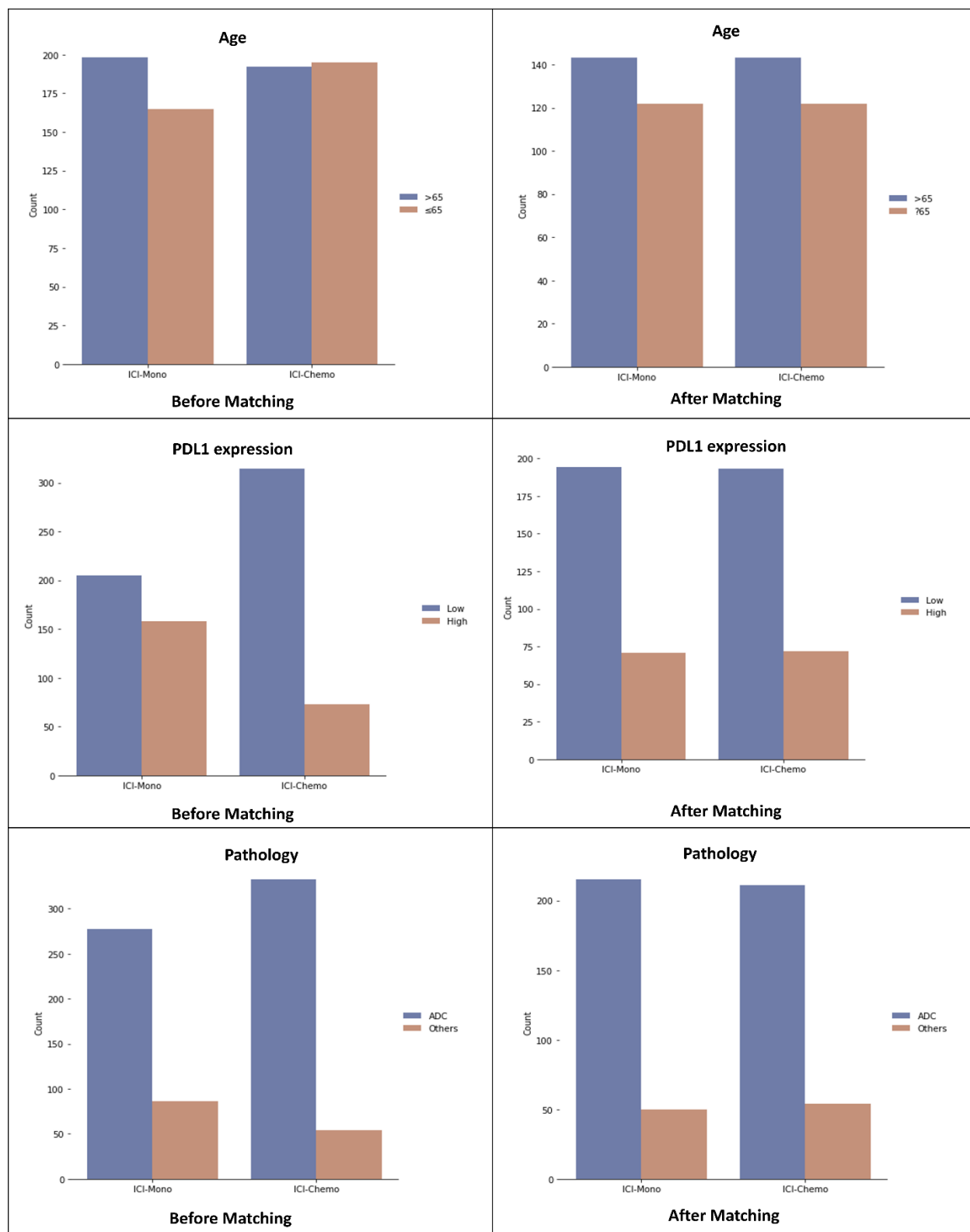


Supplementary Fig S9: Interaction plots for sub-models in external validation cohort

A few minor comments:

- For propensity score matching, it would be more customary to show the imbalance of the `_covariates_` (and not the propensity scores themselves, as done in Figure 5) before and after matching.

Thank you for advice. As suggested, the below figure shows examples of selected covariates' distributions before and after the matching (new **Supplementary Fig S2**).



Supplementary Fig S2: Example of covariates distribution before and after matching

b. There can be overlap between prognostic variables and treatment moderators, correct?

Thank you for highlighting this important consideration. Indeed, there can be overlap between prognostic variables and treatment moderators, and recognizing this overlap is crucial for the accurate interpretation of our findings. Prognostic variables are factors that predict the outcome regardless of the treatment, while treatment moderators are variables that influence the effect of the treatment on the outcome. In our study, we accounted for the potential overlap between these variables by carefully selecting and validating the covariates included in our models. We employed statistical techniques to ensure that the interaction terms and main effects were appropriately modeled to distinguish between their prognostic and moderating roles. These help to ensure that our conclusions about treatment effects are not confounded by the prognostic properties of the variables.

c. The reference to "three scoring benefits and five loss functions" in the abstract is not really understandable without having read the paper.

Thank you for this advice. The abstract has been revised accordingly following our new proposed ensemble learning as shown below.

"The recent introduction of immune checkpoint inhibitors (ICIs), either alone (ICI-Mono) or with chemotherapy (ICI-Chemo), has improved survival for patients with advanced non-small cell lung cancer (NSCLC). However, there are few data to guide which patients benefit most from ICI-Chemo as compared to ICI-Mono, and as such, further evidence is needed to refine patient selection for these approaches and further optimize individual patient outcomes. In this study, we applied artificial intelligence (machine learning) to develop a predictive model for individualizing the decision to deliver ICI-Mono or ICI-Chemo. Subjects with metastatic NSCLC were pooled from MD Anderson (n=750), Mayo Clinic (n=80), and Dana-Farber (n=1077) along with a previously published multi-institutional cohort (Stand Up To Cancer, n=393). We formulated personalized benefit scores estimating the likelihood of benefit from adding chemotherapy to ICIs using five distinct weighting-based scoring functions based on 28 genomic and 6 clinical predictors. An integrated model was then developed to model heterogeneous treatment effects between ICI-Mono and ICI-Chemo for patients at risk for early progression. Evaluation was reported as average treatment effect in terms of weighted risk reduction (WRR) for 3-month progression between patients who were treated according to the model's recommendation vs. against the model's recommendation. Overall, the model recommended over half of the patients to switch treatments (53% in discovery 50% in external validation), with three-fourths of these patients expected to benefit from adding chemotherapy. The integrated model (Attention-based Scoring for Treatment Effect Prediction, A-STEP) achieved the best risk reduction, ranging from -13% to -23% in comparison to the stand-alone scoring models. When stratified by actual treatment plans, patients treated according to A-STEP's recommendation had longer 2-year PFS as validated in the external cohort, with hazard ratio [HR, anti-subgroup as reference] = 0.60, p=0.016 in the ICI-Mono arm and HR=0.58, p=0.0076 in the ICI-Chemo arm. Notably, patients who received chemotherapy as suggested by A-STEP had 4-5 months longer median survival. Additionally, the interplay between genomic markers such as FBXW7 and APC along with PD-L1 expression were associated with potential benefits from combining immunotherapy and chemotherapy. These methods and findings offer valuable insights for integrating precision medicine into clinical practice, enhancing the potential for personalized therapeutic strategies and improved patient outcomes."

d. Is figure 1 really needed? Personalized treatment is not really a novel concept anymore these days.

Thank you for your constructive comment. We agree that the concept of personalized treatment is widely recognized in the current medical and scientific community. Given this, the previous **Figure 1**, which provides a general overview of personalized treatment, is indeed redundant. In response to your suggestion, we have replaced it with a **new Figure 1** to streamline the presentation of the overall study design and highlight the novel aspects of our research. Thank you once again for your valuable feedback, which has significantly contributed to the improvement of our manuscript.

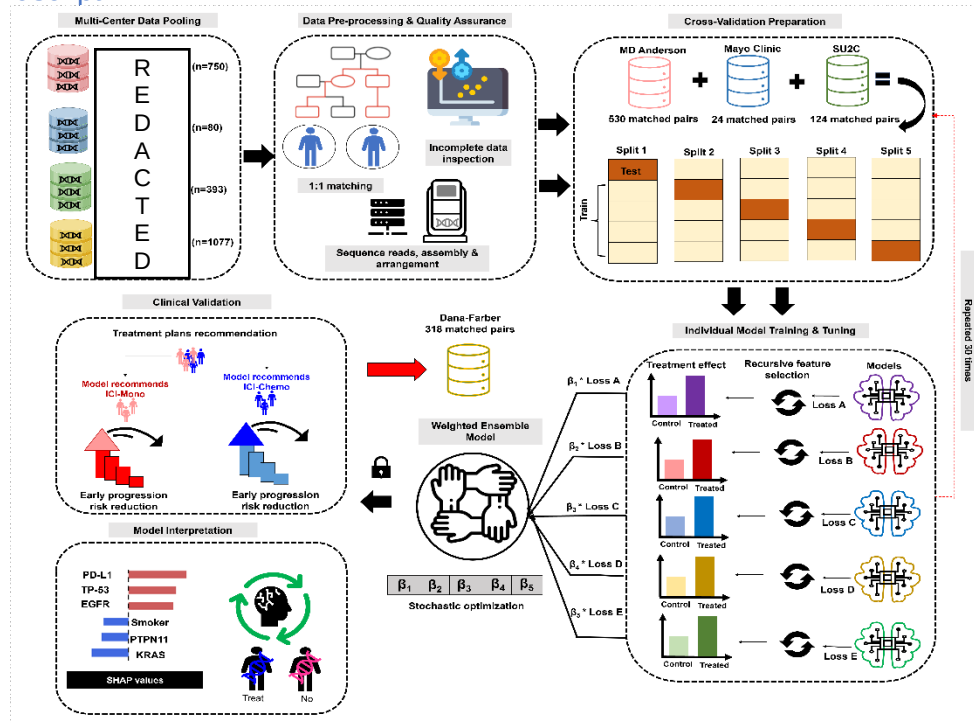


Figure 1: A-STEP diagram

e. The authors choose to identify 3-months progression as a binary endpoint. I wonder why this choice was made rather than using the full survival data, since the R package "personalized" supports time-to-event outcomes.

Thank you for your insightful question. In the era of immunotherapy, who will need chemotherapy to anti-PD1/PD-L1 ICIs is one of the most important questions. Although there is not a definitive answer to this question yet, previous studies have demonstrated that the patients who may benefit from adding chemotherapy are the early progressors. Therefore, 3-month progression is an early on-treatment assessment point during management, when it is critical to adjust treatment regimen. Thus, early progression such as 3-month progression is considered as a clinically meaningful endpoint, providing a clear and immediate measure of treatment impact. In addition, we have added the evaluation of our predictive model regarding long-term survival stratification under different treatment arms. The model remains predictive as shown in the new **Fig 3**.

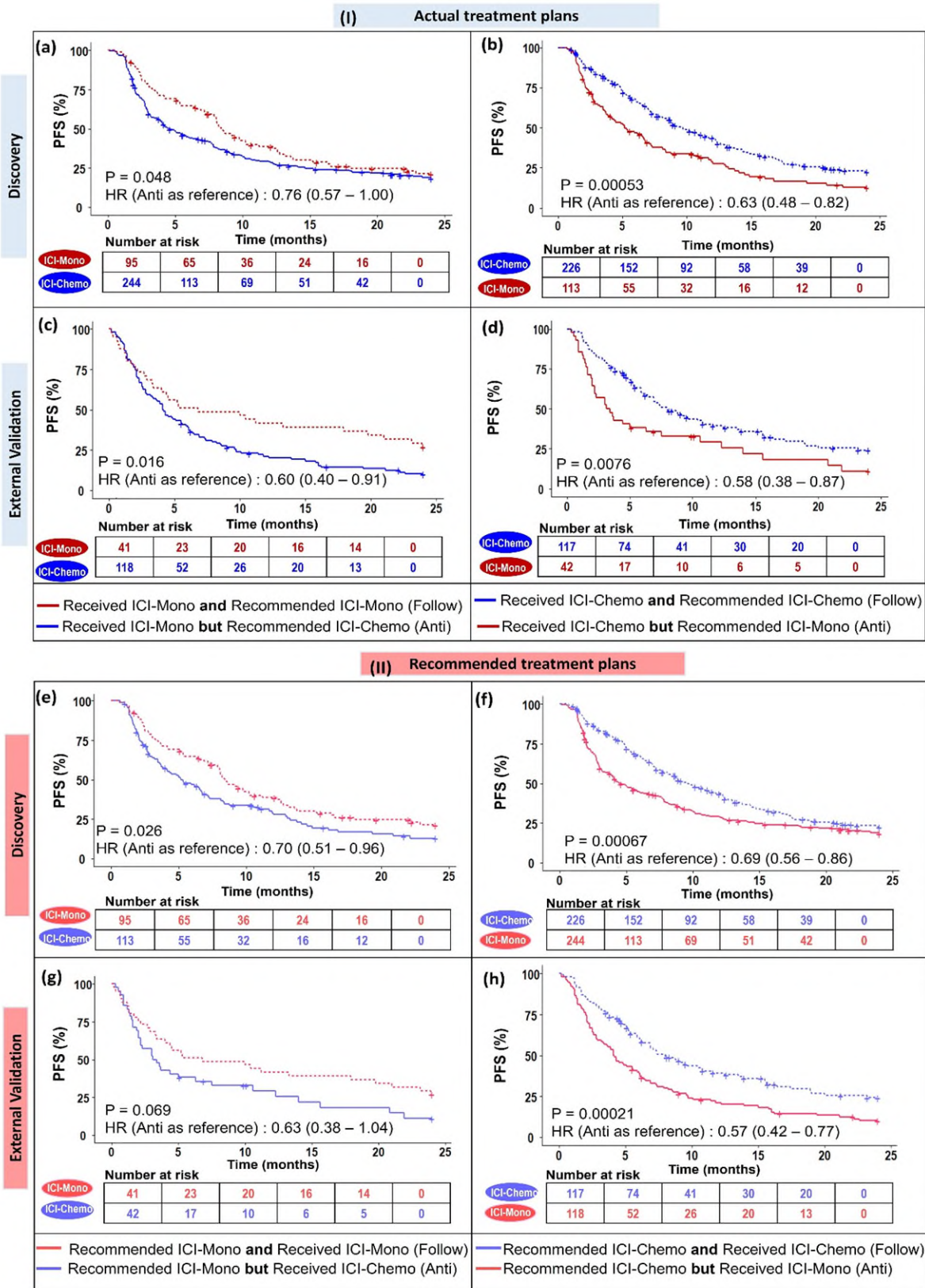


Figure 3: Comparison of 2-year PFS stratified by (I) actual treatment plans and (II) recommended treatment plans by the A-STEP

f. Are the p-values reported in Table 1 for the original unmatched MDACC cohort? (E.g there is no apparent difference between Line of ICI but the p-value of 0.002 is very low)

This is a good catch. We have now corrected and updated our Table 1, with the addition of the new samples. Please see updated **Table 1**.

Table 1: Patient demographics

Characteristic	Discovery cohort		p#	External validation cohort		p#	P**
	ICI-mono (N=339)	ICI-chemo (N=339)		ICI-mono (N=159)	ICI-chemo (N=159)		
Age, N (%)			0.113			0.064	
>65 years	180 (53.1)	179 (52.8)		78 (49.1)	87 (54.7)		0.688
≤65 years	159 (46.9)	160 (47.2)		81 (50.9)	72 (45.3)		
Gender, N (%)			0.568			0.252	
Male	186 (54.9)	177 (52.2)		84 (52.8)	85 (53.5)		0.781
Female	153 (45.3)	162 (47.8)		75 (47.2)	74 (46.5)		
Smoker, N (%)			0.274			0.191	
Yes	276 (81.4)	271 (79.9)		120 (75.5)	113 (71.1)		0.034
No	63 (18.6)	68 (20.1)		39 (24.5)	46 (28.9)		
Pathology, N (%)			0.481			0.045	
LUAD	279 (82.3)	272 (80.2)		134 (84.3)	120 (75.5)		0.620
LUSC	48 (14.2)	52 (15.3)		17 (10.7)	22 (13.8)		
Others	12 (3.5)	15 (4.4)		8 (5.03)	17 (10.7)		
Line of ICI, N (%)			0.003			0.446	0.023
1 st	197 (58.1)	263 (77.6)		70 (44.0)	144 (90.6)		
2 nd	114 (33.6)	43 (12.7)		67 (42.1)	11 (6.9)		
Others	28 (8.3)	33 (9.7)		22 (13.9)	4 (2.5)		
PD-L1 TPS, N (%)			0.398			0.012	
< 1%	133 (39.2)	126 (37.2)		72 (45.3)	72 (45.3)		0.665
1-49%	117 (34.5)	122 (36.0)		51 (32.1)	38 (23.9)		
≥ 50%	89 (26.3)	91 (26.8)		36 (22.6)	49 (30.8)		

* TPS: Tumor proportion score

Comparison between patients from two treatment arms within a cohort

** Comparison between discovery and testing cohorts

g. The choice of loss function quite dramatically impacts how many people are selected for each treatment. For instance, with the linear model, the majority are recommended to receive ICI-Chemo.

We acknowledge the uncertainties associated with the choice of different loss functions. The majority of weight-based models are skewed by recommending patients to receive ICI-Chemo. This can be affected by the cut off value used in this study, which is at 0. This cutoff means that, even when a system detects slight potential benefit of adding chemotherapy, it will recommend adding it to immunotherapy. To directly address such uncertainty, we took an effort to introduce an ensemble learning model to capture common and strong signals present across these different

loss functions while reducing individual bias and variance. Our investigation revealed that the integrated models outperformed the standalone models in terms of its performance and seems to slightly remedy the issue of the system bias towards ICI-Chemo (new **Table 3**). For example, the best stand-alone model (sub-model B) only recommends that 9.7% of patients receive ICI-Mono, while the best model (A-STEP) recommends up to 26% of patients receive ICI-Mono.

Table 3: Refitted treatment effects reported as weighted risk reduction (WRR) in all sub-scoring and integrated models. Negative sign represented reductions in risk.

Model	Discovery		External Validation	
	ICI-Mono (%)	ICI-Chemo (%)	ICI-Mono (%)	ICI-Chemo (%)
A	-26.5 (n=54)	-15.1 (n=624)	13.6 (n=29)	-9.2 (n=289)
B	-3.9 (n=66)	-13.8 (n=612)	-9.2 (n=31)	-12.1 (n=287)
C	-26.5 (n=54)	-15.1 (n=624)	13.6 (n=29)	-9.2 (n=289)
D	-21.5 (n=75)	-16.2 (n=603)	21.4 (n=34)	-18.4 (n=284)
E	1.1 (n=315)	-20.1 (n=363)	-5.1 (n=115)	-18.8 (n=203)
H-STEP	4.0 (n=366)	-21.5 (n=312)	0.7 (n=142)	-18.4 (n=176)
S-STEP	-29.6 (n=56)	-15.7 (n=622)	2.2 (n=32)	-9.9 (n=286)
A-STEP	-13.5 (n=208)	-23.4 (n=470)	-16.0 (n=83)	-19.3 (n=235)

- Reviewer #2 (Remarks on code availability): The authors stated that they will only make the code available upon publication (p. 25) so I was unable to review the code.

Thank you for your feedback regarding the availability of our codes. We appreciate the importance of transparency and reproducibility in scientific research. We have included the codes in this revised manuscript to ensure that it is available for review. The codes are now accessible and can be reviewed alongside the manuscript. Additionally, we have made the codes publicly available in a dedicated repository on [<https://github.com/WuLabMDA/A-STEP>], with detailed documentation to facilitate its use and replication of our results. Thank you once again for your valuable feedback.

Reviewer #3

The overarching goal of this paper is to improve treatment selection for individual patients using baseline covariates. Specifically, the clinical objectives of this study are to assess the impact of adding chemotherapy on early progression outcomes in NSCLC patients receiving ICI therapy. The paper is well written and the authors utilize many elements of modern statistical practice including the use of propensity score matching; the use of a discovery sample for model development and an external-validating testing sample for assessing how well the models do with out-of-sample predictions, i.e. subgroup identification; and the use of sensitivity analysis, including several different model specifications for loss functions.

An issue for this reader with respect to the field of personalized medicine in general is the use of terminology, such as “personalized immunotherapy” or the expression of an objective to seek “optimal personalized treatment strategies” implying that the outcome of this work will be the identification of the optimal treatment for an individual. In fact, as the authors rightly note (p.3: L 88) “The process of determining the best treatment allocation can be viewed as a task of subgroup identification, wherein subgroups are identified based on the heterogeneity in the treatment effects.” What is achievable statistically is to identify a sub-group of patients who share common characteristics and investigate what treatment works best for such subgroups. In other, words the “optimal personalized treatment strategy” is the treatment for the canonical individual who best represents a sub-group.

The technological sophistication of this work is quite high consisting of multiple components. One element of this program is the specification of different loss functions. The authors highlight that (p. 11: L248) *“The analysis shows the influence of various loss functions on predictive biomarker identification for early progression in patient cohorts undergoing ICI-Mono or ICI-Chemo.”* See figure 6. They write (p.12: L275): *“This investigation yielded a striking revelation: There are a rather limited number of overlapping features among the diverse loss functions, strongly suggesting that the distinct loss functions are effectively capturing discrete information or intricate patterns present within the data. This inherent diversity in the loss functions’ capabilities resulted in the identification of distinct sets of biomarkers, each carrying its own valuable insights.”*

One wonders whether this is in fact the only explanation and whether this result is what the authors expected. Would the use of other loss functions have identified other sets of biomarkers that would also carry valuable insights? From the point-of-view of sensitivity analysis wouldn't one have hoped that the results were robust to the specification of different loss functions? In the end they do report a singular predictive biomarker – the mutated variant EGFR, that impacts treatment response for patients treated with ICI-Chemo, whereas among patients undergoing ICI-mono treatment, the use of tobacco emerged as a predictive factor. One wonders if these predictive factors could have been identified using less sophisticated methodologies? In other words, are there alternative approaches for identifying patients prone to early progression within treatment groups and the biomarkers that are associated with treatment response?

We thank the reviewer for highlighting the complexity and critical aspects regarding how different scoring functions identify different sets of biomarkers. We agree with your inference that had we explored more scoring functions, it could have led to the detection of a different set of biomarkers. This observation, however, is not unexpected given the fact that individual techniques are inherently diverse in their capabilities of detecting various patterns and interactions (i.e. there is no one-size-fits-all model). To directly address this uncertainty, we have introduced an ensemble learning mechanism into the study design that takes advantage of different integration algorithms (**Suppl Method S4**). This helps uncover the strong common and potentially complimentary

signals across distinct scoring functions. The new **Table 3** shows the three ensemble models that we investigated (H-STEP, S-STEP, and A-STEP), where A-STEP (i.e., the attention-based integration model) demonstrated the most robust performance.

Table 3: Refitted treatment effects reported as weighted risk reduction (WRR) in all sub-scoring and integrated models. Negative sign represented reductions in risk.

Model	Discovery		External Validation	
	ICI-Mono (%)	ICI-Chemo (%)	ICI-Mono (%)	ICI-Chemo (%)
A	-26.5 (n=54)	-15.1 (n=624)	13.6 (n=29)	-9.2 (n=289)
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D	-21.5 (n=75)	-16.2 (n=603)	21.4 (n=34)	-18.4 (n=284)
E	1.1 (n=315)	-20.1 (n=363)	-5.1 (n=115)	-18.8 (n=203)
H-STEP	4.0 (n=366)	-21.5 (n=312)	0.7 (n=142)	-18.4 (n=176)
S-STEP	-29.6 (n=56)	-15.7 (n=622)	2.2 (n=32)	-9.9 (n=286)
A-STEP	-13.5 (n=208)	-23.4 (n=470)	-16.0 (n=83)	-19.3 (n=235)

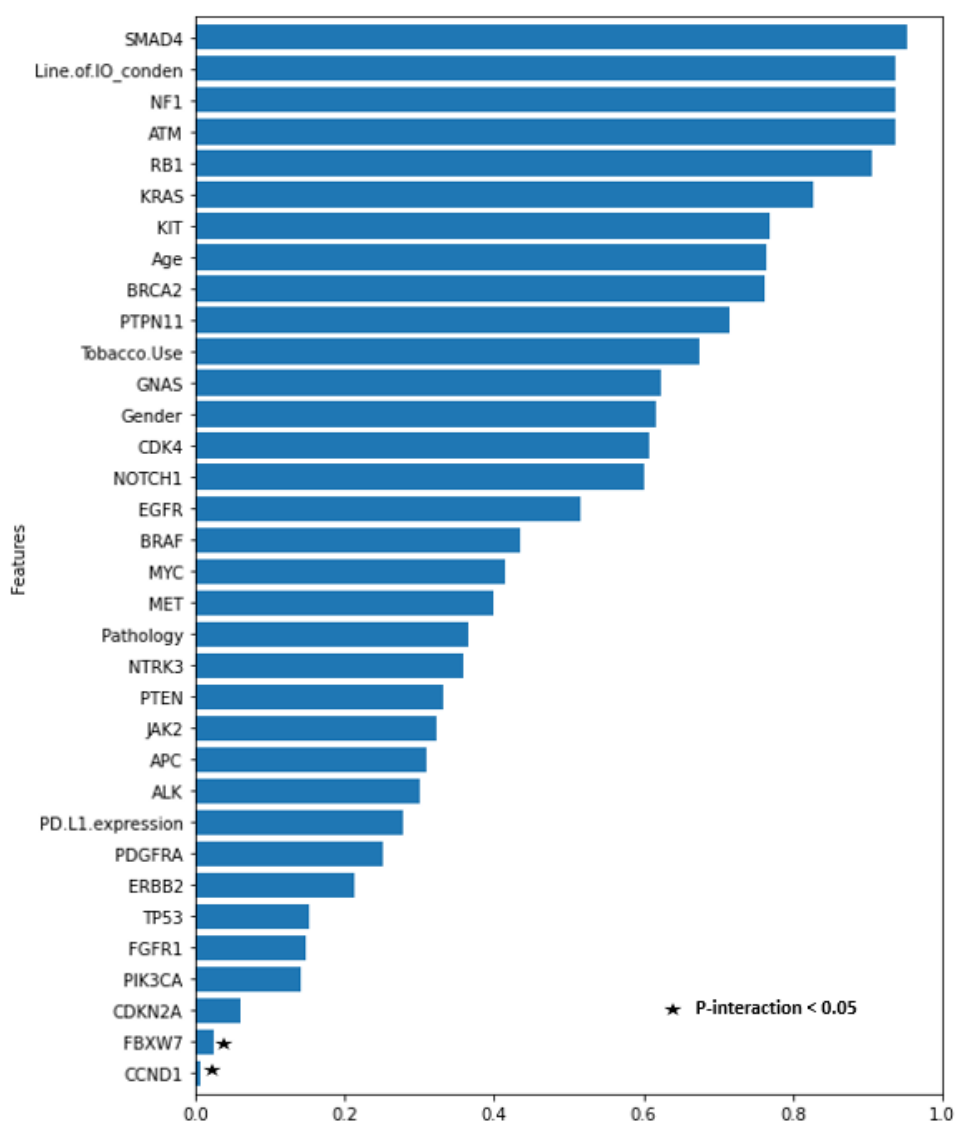
It is also worth mentioning that these models are complex and highly non-linear in nature, driven by the interplay across different features. Their stratification is not governed by a singular feature, as observed in our univariate analyses (**Supplementary Fig. S15**).

Additionally, addressing the reviewer's question on the possibility of using lower complexity modeling approaches to identify and develop predictive biomarkers, we have investigated two conventional strategies available in the literature: (1) one-at-a-time analysis and (2) risk-based analysis:

One-at-a-time analysis

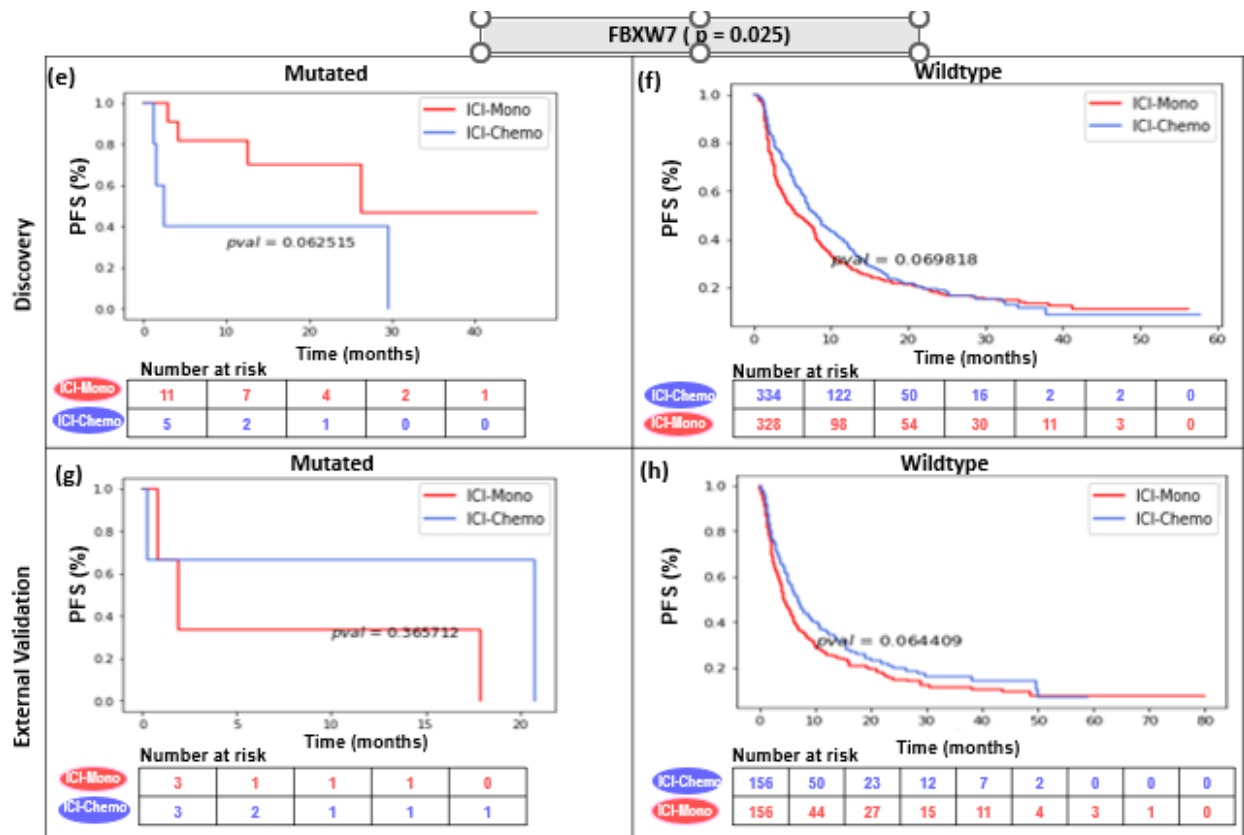
- First estimate the interaction between treatment indicator and individual covariate through a regression model
- Retain covariates with p-interaction < 0.05, meaning they have potential to inform treatment effect.
- Then perform simple subgroup analyses where the treatment and control arms are compared in different subgroups defined by identified individual covariate.

The new **Supplementary Fig S10** demonstrates the treatment-covariate interaction measured for all features in this study. Only two genomic features were found to be predictive (p-interaction<0.05), which were **FBXW7** and **CCND1**.



Supplementary Fig S10: Treatment-covariate interaction

The new **Supplementary Fig S11** shows the experiment of the **FBXW7** genomic predictor, with p-interaction = 0.025, yet failed to stratify patients for treatment efficacy, especially in the external validation cohort. Another issue is that very few patients were identified with that mutation, with 16 out of 678 in discovery and 6 out of 318 in external validation.

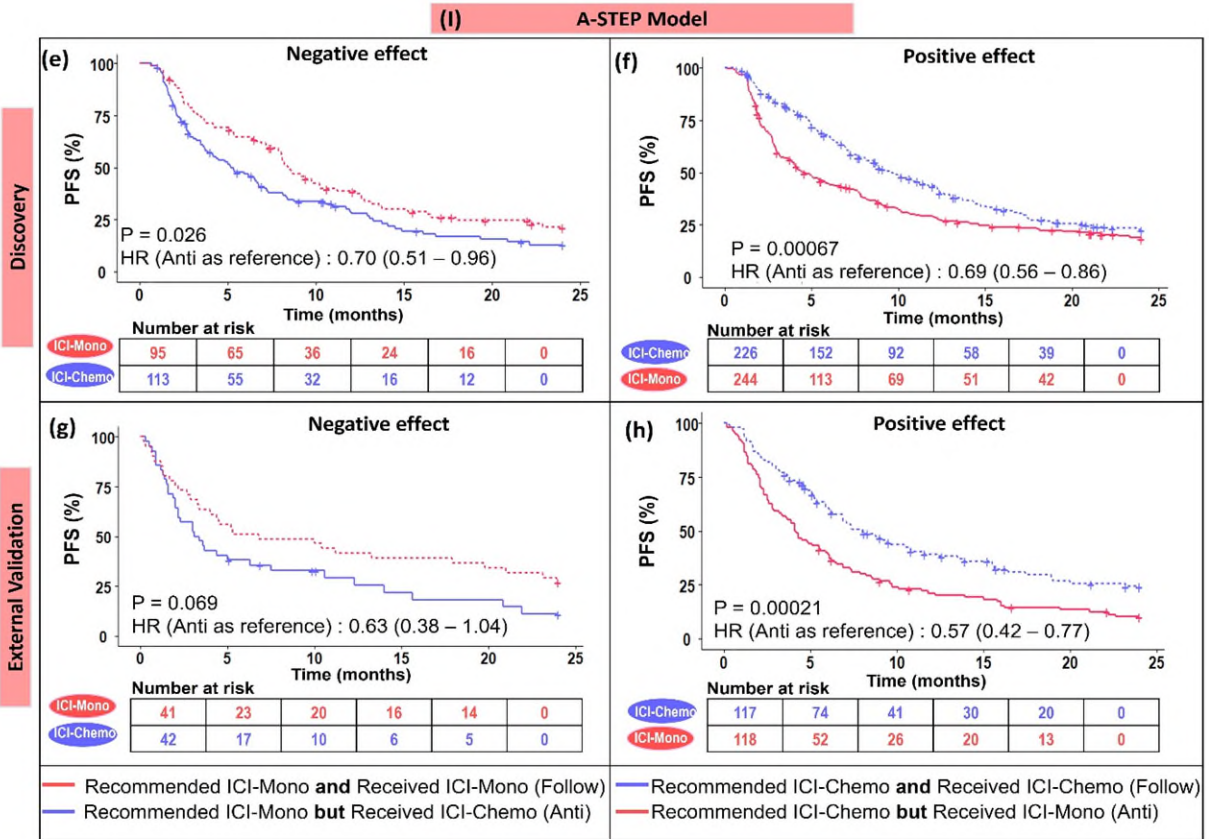


Supplementary Fig S11: One at a time analysis

Risk-based analysis

- Build a prognostic model using clinicogenomic features and utilize the risk scores to stratify patients into low and high-risk groups.
- Then perform subgroup analyses where the treatment and control arms are compared in low- and high-risk groups.

New **Supplementary Fig S12** demonstrates the comparison between the proposed A-STEP model vs. the risk-based method. It was observed that the risk-based method also failed to act as a predictive model according to the standard as defined here (<https://ascopubs.org/doi/10.1200/JCO.2015.63.3651>). Taken together, these additional experiments suggest that simpler modeling fails to reach the same level of performance.



Reviewer #4

1. The authors have devised an approach to selecting patients for optimal treatment choice. This leverages various models to determine the 'individualized treatment rules' for each patient, to find whether they would derive most benefit from ICI (an acronym they do not define in the abstract) or chemo/ICI. They leverage a 750 patient cohort and validate their approach in a further 80 patients. Unfortunately, it is of the Reviewer's opinion that this manuscript should be rejected.

Thank you for your comprehensive review of our manuscript.

We have worked very hard to now significantly improve the overall design of this study based on your and other reviewers' constructive comments. This study aims to develop an innovative computational predictive framework and demonstrate its potential impact on a relevant clinical decision, focusing on the selection between ICI-mono vs ICI-chemo. We appreciate the reviewer's constructive comments and wish to highlight a few major changes we have made to address the concerns raised: (1) an ensemble learning scheme was used to combine different weighting methods, aiming to capture common and strong signals present across these different algorithms while reducing individual bias and variance, (2) two additional datasets including Dana-Farber Cancer Institute (n=1077) along with one multi-site consortium from Stand Up To Cancer (n=393) were added to develop and validate algorithms, and (3) beyond predicting early disease progression status, the model's predictive performance was rigorously assessed based on the full progression free survival data, (4) the quality of figures and tables were significantly improved, with 4 new figures (Figure 1-4), 2 new tables (Table 2-3), 15 new supplementary figures (Suppl Fig S1-S15), and 1 new supplementary table (Suppl Table S1). Further, we acknowledge the limitations of this study, and clarified that our study is a proof-of-principle study, which needs future validation in future studies before clinical translation.

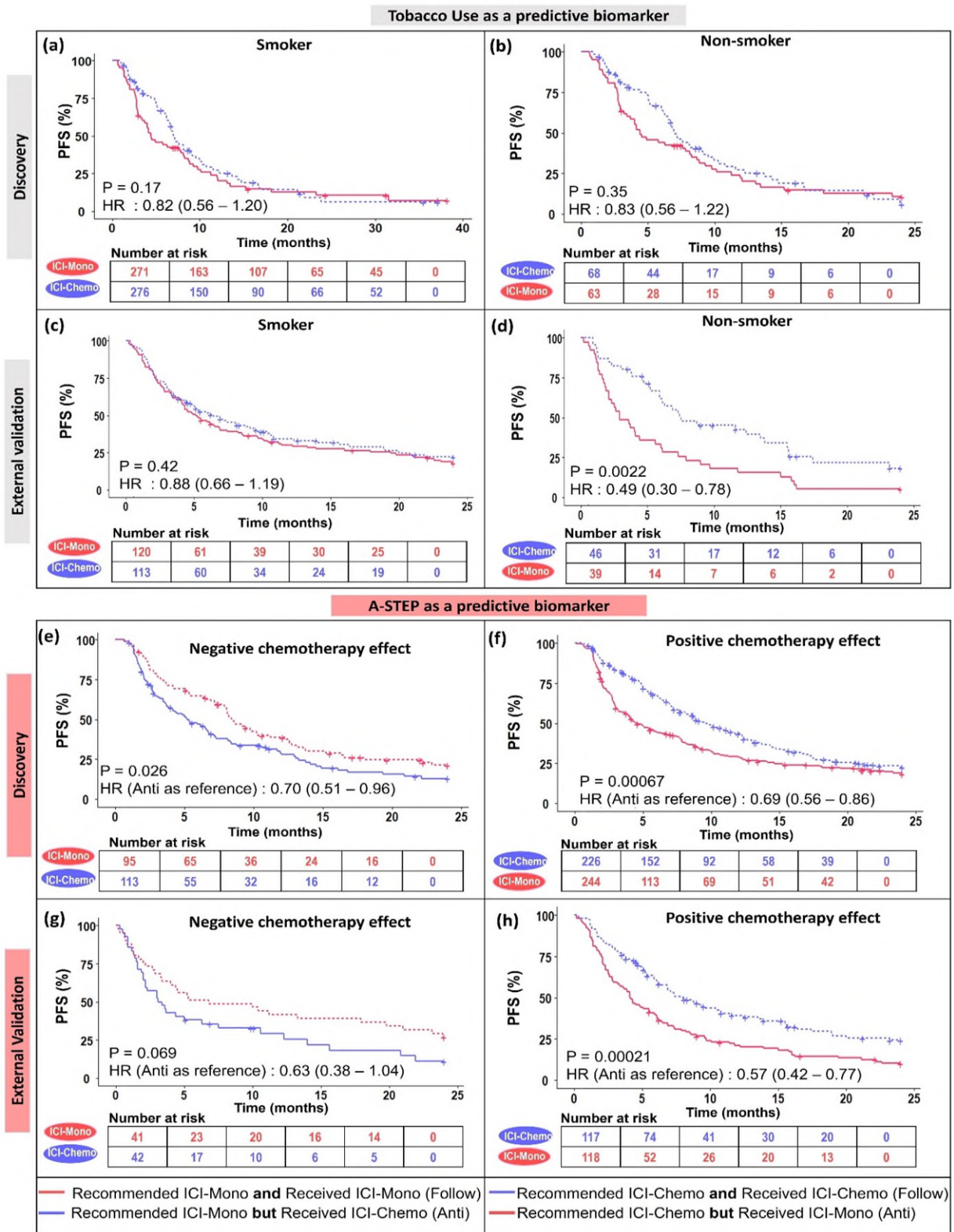
We have also now defined ICI (Immune Checkpoint Inhibitor) in the abstract, please see appended.

"The recent introduction of immune checkpoint inhibitors (ICIs), either alone (ICI-Mono) or with chemotherapy (ICI-Chemo), has improved survival for patients with advanced non-small cell lung cancer (NSCLC). However, there are few data to guide which patients benefit most from ICI-Chemo as compared to ICI-Mono, and as such, further evidence is needed to refine patient selection for these approaches and further optimize individual patient outcomes. In this study, we applied artificial intelligence (machine learning) to develop a predictive model for individualizing the decision to deliver ICI-Mono or ICI-Chemo. Subjects with metastatic NSCLC were pooled from MD Anderson (n=750), Mayo Clinic (n=80), and Dana-Farber (n=1077) along with a previously published multi-institutional cohort (Stand Up To Cancer, n=393). We formulated personalized benefit scores estimating the likelihood of benefit from adding chemotherapy to ICIs using five distinct weighting-based scoring functions based on 28 genomic and 6 clinical predictors. An integrated model was then developed to model heterogeneous treatment effects between ICI-Mono and ICI-Chemo for patients at risk for early progression. Evaluation was reported as average treatment effect in terms of weighted risk reduction (WRR) for 3-month progression between patients who were treated according to the model's recommendation vs. against the model's recommendation. Overall, the model recommended over half of the patients to switch treatments (53% in discovery and 50%

in external validation), with three-fourths of these patients expected to benefit from adding chemotherapy. The integrated model (Attention-based Scoring for Treatment Effect Prediction, A-STEP) achieved the best risk reduction, ranging from -13% to -23% in comparison to the stand-alone scoring models. When stratified by actual treatment plans, patients treated according to A-STEP's recommendation had longer 2-year PFS as validated in the external cohort, with hazard ratio [HR, anti-subgroup as reference] = 0.60, $p=0.016$ in the ICI-Mono arm and $HR=0.58$, $p=0.0076$ in the ICI-Chemo arm. Notably, patients who received chemotherapy as suggested by A-STEP had 4-5 months longer median survival. Additionally, the interplay between genomic markers such as FBXW7 and APC along with PD-L1 expression were associated with potential benefits from combining immunotherapy and chemotherapy. These methods and findings offer valuable insights for integrating precision medicine into clinical practice, enhancing the potential for personalized therapeutic strategies and improved patient outcomes."

2. The observation that smokers appear to benefit more from ICI-mono than ICI-chemo is moderately interesting. They do not validate this observation, instead focusing on validating their approach for patient selection.

Thank you for bringing up this point. We agree that the observed benefit from adding chemotherapy to never smokers is an important observation. Following your advice, the new **Supplementary Fig S13** displays the subgroup analyses between smokers and never smokers stratified by treatment arms. We observed that non-smokers benefit from adding chemotherapy, but there was no significant difference when it comes to smokers. However, this trend fluctuates between different cohorts. For the comparison purposes, we plotted the proposed method as presented in figure below, which has consistent predictive performance.



We also note that smoking status as a feature that associates with differential benefit to ICI-Mono vs ICI-Chemo is consistent with previous reports, including previously published meta-analyses and real-world data [doi: [10.1093/jnci/djab116](https://doi.org/10.1093/jnci/djab116) and <https://doi.org/10.1080/0284186X.2019.1670354>]. This external validation highlights the capability of the proposed models to identify clinically relevant predictive features and suggests that novel features that are highlighted may also be worth further investigation and validation (Fig 5).

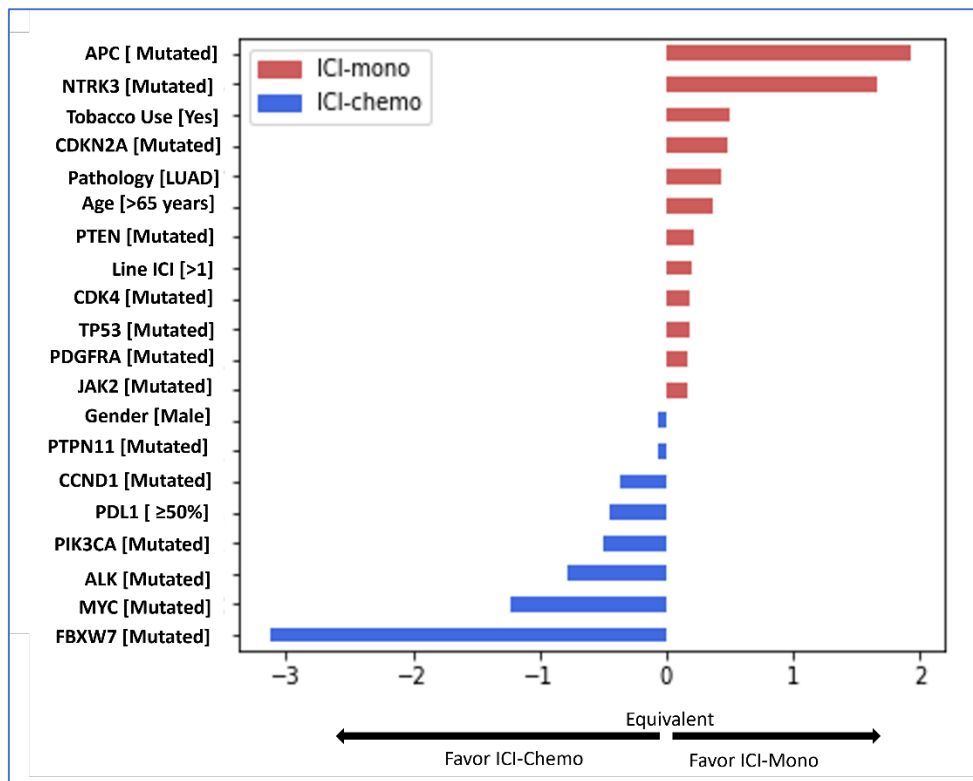


Figure 5: A-STEP SHAP interpretation.

- As their Figure 1 demonstrates, for precision medicine to be effective, there must be multiple theoretical options available that, if given, would provide a patient with meaningful clinical benefit. This is not the case in this (or any) indication. They present this problem as the focus of ‘a substantial effort with the statistics and machine learning community’ when really, it is a basic science / drug discovery / drug development / trials / data availability problem (even without acknowledging the problem of pre-existing or acquired resistance).

Thank you for your valuable feedback on our manuscript. We appreciate your insights and agree that for precision medicine to be truly effective and needed, there must be multiple effective therapeutic options to select between. Our study is not intended to solve the basic science/ drug development challenges of precision treatment directly, as you correctly pointed out. Our research serves as a proof-of-principle demonstration of how machine learning approaches can be leveraged to address these complex clinical decisions as they arise, particularly in contexts where treatment efficacy cannot be linked to a simple e.g. genomic biomarker. Further, while there is much work to be done to develop better therapies, it is also

the case that even now, providers are being faced with the clinical question of choosing between treatment regimens that have different, even if modest, efficacy. Specifically, we used the ICI-Mono vs ICI-Chemo question as an illustrative example to showcase the potential utility of our approach in guiding treatment decisions, particularly in situations where clinicians face a lack of head-to-head trial data.

We understand and agree that our initial manuscript may have overstated the immediate clinical applicability of our findings. To address this, we have revised the manuscript to clarify that our aim is to complement ongoing drug discovery efforts by providing a statistical and machine learning framework that can help prioritize and evaluate potential treatment options based on patient-specific data. We recognize that our work is one piece of a larger puzzle and does not negate the critical role of basic science, drug development, and clinical trials.

In response to your comments, we have also added the following to the discussion section:

"Finally, while the development of multiple therapeutic options is inherently a challenge of basic science, drug discovery, and clinical trials, our approach aims to complement these advances by utilizing statistical and machine learning models to identify the most promising treatment strategies for individual patients with the treatments already available. We recognize the critical role of these foundational efforts and aim to build upon them to enhance the precision and effectiveness of clinical decision-making."

We hope these revisions address your concerns and demonstrate our commitment to enhancing the quality and clarity of our manuscript. We greatly appreciate your constructive feedback and look forward to your further input.

4. Therefore, the novelty here must come from the uniqueness of the modelling framework they propose, or the data they present. Regrettably, the data is of limited utility, as they are not proposing sharing it (available from the corresponding author 'upon request'), they don't define what it is in the Methods (was it WGS/WES/a panel?), and the modelling framework is very opaque, poorly explained and with no code available to review, and of dubious/no benefit.

Thank you for highlighting these critiques. There are four overarching concerns expressed in this comment which we have attempted to address in turn:

Data Availability and Utility: We apologize for any confusion regarding the accessibility and utility of our data. We have revised our manuscript to clearly state that the data will be made available in a public repository. This will allow other researchers to access and utilize the data more readily. We have clarified this in the revised Data Availability Statement: "The data used in this study will be deposited in [Zenodo: 10.5281/zenodo.13368111] and will be accessible to the research community upon reasonable request. This ensures that the data can be independently verified and used for further research."

Clarification of Genotype Data: We have added detailed descriptions of the data types used in our study, which is from CLIA-certified, standard-of-care targeted panels.

Transparency and Explanation of the Modeling Framework: We recognize that the original manuscript did not sufficiently explain the modeling framework. We have significantly expanded this section to provide a clear and detailed description of the framework, including the algorithms used, parameter settings, and the rationale behind our approach. Additionally, we have made the codes available in a public repository for review and replication. The

complete codes and detailed documentation are available at [\[https://github.com/WuLabMDA/A-STEP\]](https://github.com/WuLabMDA/A-STEP).

Novelty – Ensemble learning scheme: we have proposed a new attention-based mechanism to combine different weighting methods, which proved to be an effective framework to capture common and strong signals present across these different algorithms while reducing individual bias and variance. **Assessment of Benefit on Large External Cohorts:** To address the concern about the potential benefit of our framework, we have included additional datasets to demonstrate the efficacy and advantages of our approach. These results highlight the clinical relevance and potential impact of our work.

We hope these revisions address your concerns and enhance the clarity and robustness of our manuscript. We are committed to ensuring the highest standards of transparency and utility in our research. Thank you again for your constructive feedback and for helping us improve our work.

Further points:

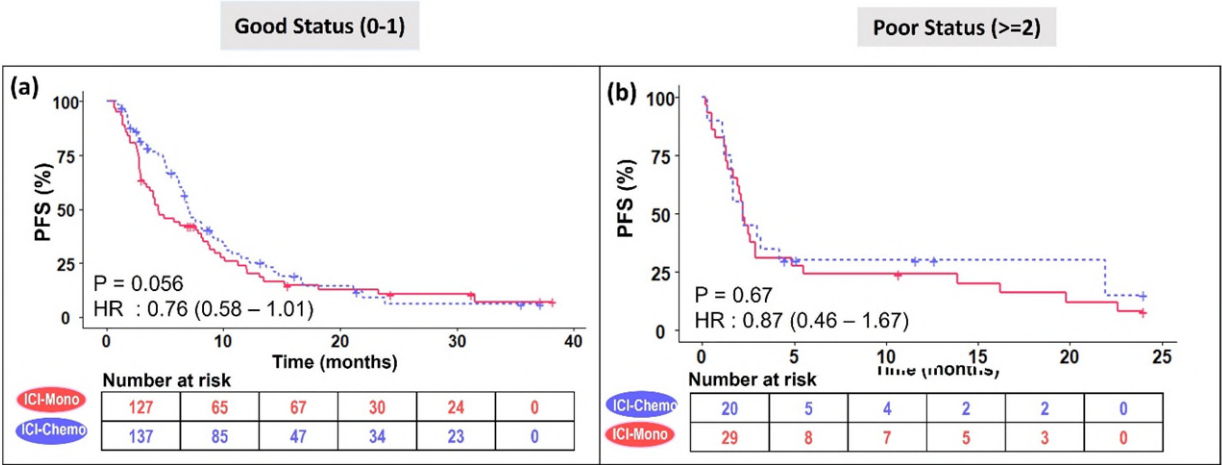
- a. Their study hinges on their ability to identify and control for bias arising from intrinsic differences between patients selected for ICI-mono and ICI-chemo – the reviewer is not convinced that with their limited clinical data, that does not for example include Performance Status, they would be able to do this meaningfully.

Thank you for raising this important concern. We completely agree that the real-world retrospective data contains intrinsic biases that can only be partially overcome, and that there may be important factors that influence patient outcome that we may not have captured in these large, real-world clinical cohorts. Regarding the specific concern about performance status, we did not include performance status in our analyses because ECOG assessments from the real-world medical records were highly skewed to ECOG of 0-1 (~ 70% of patients) such that discussed in Sorensen et al Br J Cancer 1993;67:773-775 [doi: [10.1038/bjc.1993.140](https://doi.org/10.1038/bjc.1993.140)] and Buccheri et al. [[https://doi.org/10.1016/0959-8049\(95\)00664-8](https://doi.org/10.1016/0959-8049(95)00664-8)]. In our case, a large proportion of the data (~20%) were unavailable/missing values, which would have caused further loss of a large number of patients without adding meaningful additional information in the majority of patients. We acknowledge that this is an important limitation in our data.

We further investigated if ECOG has the potential to impact treatment selection by performing a subgroup analysis as follows:

- First evaluate the interaction between treatment indicator and ECOG through a regression model
- Check if interaction p-value < 0.05, meaning it has the potential to inform treatment effect.
- Then perform simple subgroup analyses where the treatment and control arms are compared in two subgroups (ECOG 0-1/Good status) vs. (ECOG ≥2/Poor status). The ECOG categorization (Good vs. Poor) was done as per prior studies such as [doi:10.1001/jamanetworkopen.2020.37120] and [doi: [10.1002/cncr.32645](https://doi.org/10.1002/cncr.32645)].

Figure below demonstrated patient stratification by ECOG that has an interaction p-value of 0.605. We observed no treatment effect exist when comparing two treatment arms in the subgroups of good and poor performance status patients, which further reinforced our assumption that the overall impact was likely minimal.



Further, we agree that controlling for intrinsic differences between patient groups to ensure the validity of our findings is of paramount importance in this kind of analysis. While we acknowledge that our heterogeneous real-world data does not include all clinical variables, we have taken several steps to mitigate the potential impact of this limitation: 1) incorporation of multiple large cohorts from multiple sites to mitigate bias from single-center data; 2) inclusion of a range of relevant clinicogenomic variables, which we have used to adjust for potential confounders; 3) rigorous statistical techniques including propensity score matching to adjust for potential biases; 4) implementing the weighting algorithms to investigate the interactions of these features under different machine learning models, and subsequently ensemble these individual models to improve the predictive performance. With these approaches, we observed highly concordant results across cohorts, and we also note that our findings are concordant with other publications from different datasets, highlighting the reproducibility of our findings across different datasets with likely distinct intrinsic biases.

- b. There is no description of how the genomic mutation markers were derived, or what they were, anywhere in the paper.

We have clarified this in the Methods section as follows:

“We collected somatic sequencing results from genomic profiling or external vendors from pathology reports and clinical notes. Samples with in-house somatic sequencing data were included for extended genomic analyses. Mutational profiling was conducted on formalin-fixed paraffin-embedded tumor tissue or blood samples as previously described^{36,37}. For the MD Anderson data, tissue molecular profiling used NGS-based analysis to detect mutations in 134 or 146 genes. Sequencing or circulating tumor DNA (ctDNA) was performed using the MD Anderson liquid biopsy panel (70 genes) or the Guardant360 panel (74 genes). For computational power, samples from any time point were included, and a mutational event was considered present if it was detected in tissue and/or in the blood. The sample closest to the date of ICI initiation was selected for patients with multiple samples across different time points. For consistency in analysis across

sequencing panels, we limited the genomics set to the 70 genes contained in the ctDNA panel. All reported non-synonymous mutational events were included; copy number alterations with log2 copy ratio <1 (deletion) or >5 (amplification) were included in the analysis. Tumor mutational burden was excluded due to smaller sequencing panel size. Further details are described elsewhere⁹. Similar genomic event filtering was used in the Mayo, SU2C, and DFCI datasets. In the end, 28 overlapping genes across different cohorts were kept for downstream modeling analysis."

- c. This paper does not consider the possibility, which would be supported by the related literature, that they are testing the wrong covariates and that none of the markers they are testing are likely to be biomarkers for treatment as they propose (I think this is likely – I would focus on ICI-related biomarkers identified in Litchfield et al's meta analysis from Cell 2021).

Thank you for this insightful comment and constructive advice. We agree that ICI biomarkers have been elusive, demonstrated by nearly 10 years of ICI biomarker analyses with predictive performance of AUC ~0.7, and are unlikely to be fully captured by the clinicogenomic data available in routine clinical practice. Indeed, as Litchfield et al demonstrated, it is highly likely that even whole exome sequencing data does not fully capture the biology underlying ICI response. However, we note that our study is focused less on identifying the biology underlying ICI response, and more on applying a quantitative framework to clinical decision making. We do not claim that our model offers the definitive solution to ICI (or chemotherapy response), and we agree that broader molecular profiling (including single-cell data) may ultimately better uncover the causally relevant features. However, we focused on two more short-term questions, asking 1) whether any of the clinicogenomic data that is readily available could inform the treatment selection decision clinicians currently face, and 2) whether causal inference modeling could be applied to this type of clinical question. We therefore highlight two important distinctions between this analysis and the Litchfield et al paper:

1. **Focus on Mutation Data from Clinical Gene Panels:** Our study specifically focuses on mutation data derived from clinical gene panels because these are widely used in clinical settings and are therefore available on a large enough number of patients to support the modeling strategies we employed, and offer a practical approach for clinical translation in the future if validated. Many of the features identified in Litchfield et al.'s analyses, including mutational signatures and specific copy number changes, were not available from the limited panel-based sequencing available to us, and therefore may not be as readily applicable in routine clinical practice. We hope that clinically annotated cohorts with more extensive molecular profiling grow and become more widely available, allowing us to extend our approach to include these features.
2. **Predictive vs Treatment Selection Modeling:** Our primary objective is to develop a proof-of-principle, predictive framework to guide the intensification of immunotherapy. This framework is designed to provide actionable insights that can directly influence clinical decision-making regarding treatment strategies. The previous work by Litchfield et al focused on defining features associated with ICI response in general, rather than comparing the utility of two different regimens.

In summary, while we agree that the biomarkers identified in the Litchfield et al. study are important and largely not captured in our study, our goals and methodologies necessitated a different approach. We believe that our focus on genomics data from clinical assays and the development of a predictive framework provides a unique and valuable contribution to the field as a proof-for-principle study. As more genomic and transcriptomic data become available in

more patients, we can conduct future studies that integrate these additional features to enhance the robustness and applicability of predictive models for immunotherapy.

- d. The fact that their various loss models find such divergent results from the same data makes me question their value.

We acknowledge the reviewer's concern on this. From a technical perspective, this was not unexpected, given the fact that each technique is inherently diverse in its capabilities to detect various data patterns and interactions (i.e. there is no one-size-fits-all model). To address the divergent results, we introduced the novel ensemble learning approach, by taking advantage of an attention-based mechanism in which different levels of importance ('soft' weights) were assigned to the different scoring functions. This allows the capture of the strong and complimentary signals among these distinct scoring functions. The new **Table 3** shows that the ensemble model A-STEP achieved the best performance.

Table 3: Refitted treatment effects reported as weighted risk reduction (WRR) in all sub-scoring and integrated models. Negative sign represented reductions in risk.

Model	Discovery		External Validation	
	ICI-Mono (%)	ICI-Chemo (%)	ICI-Mono (%)	ICI-Chemo (%)
A	-26.5 (n=54)	-15.1 (n=624)	13.6 (n=29)	-9.2 (n=289)
B	-3.9 (n=66)	-13.8 (n=612)	-9.2 (n=31)	-12.1 (n=287)
C	-26.5 (n=54)	-15.1 (n=624)	13.6 (n=29)	-9.2 (n=289)
D	-21.5 (n=75)	-16.2 (n=603)	21.4 (n=34)	-18.4 (n=284)
E	1.1 (n=315)	-20.1 (n=363)	-5.1 (n=115)	-18.8 (n=203)
H-STEP	4.0 (n=366)	-21.5 (n=312)	0.7 (n=142)	-18.4 (n=176)
S-STEP	-29.6 (n=56)	-15.7 (n=622)	2.2 (n=32)	-9.9 (n=286)
A-STEP	-13.5 (n=208)	-23.4 (n=470)	-16.0 (n=83)	-19.3 (n=235)

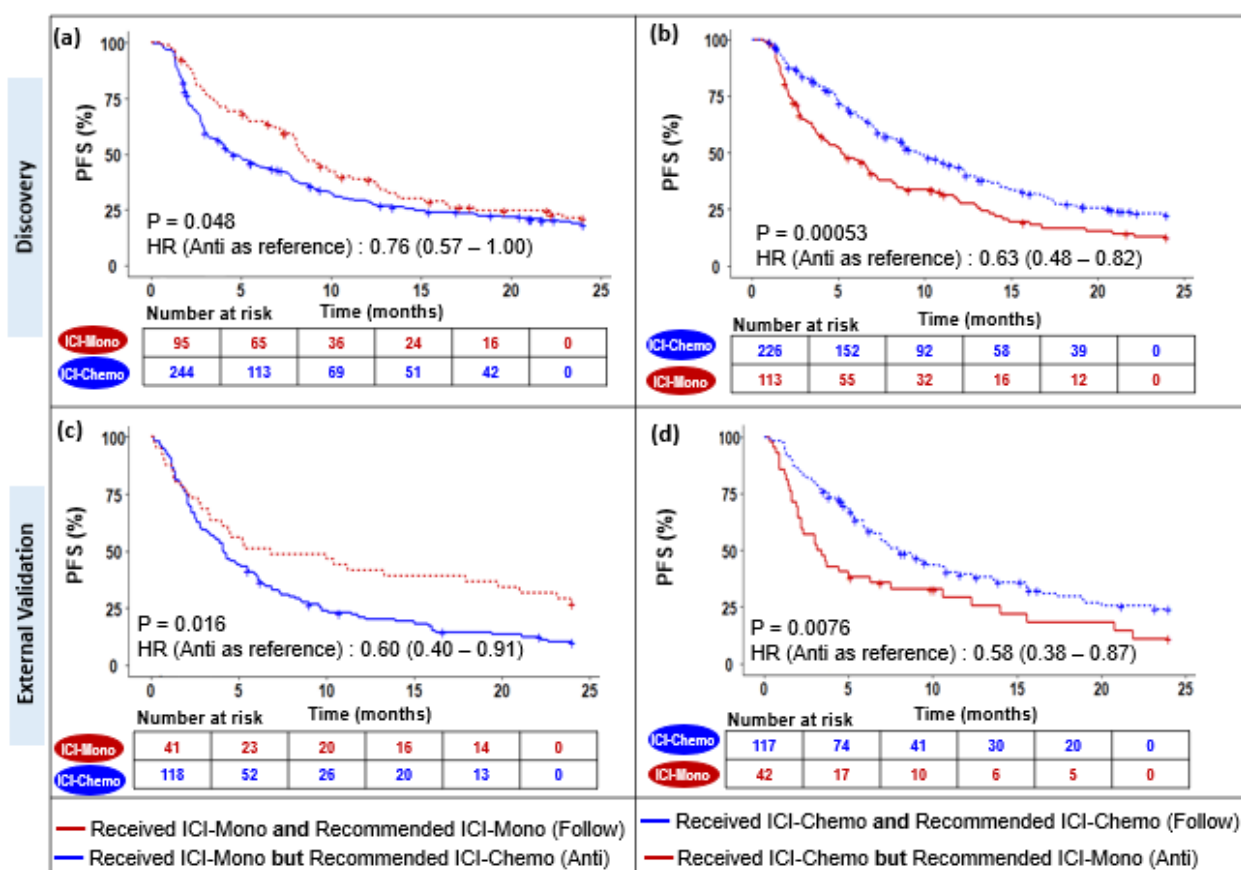
- e. There is no indication of what would constitute 'success' as far as their validation cohort is concerned. I could not ascertain a p value or understand how much improvement one might expect by chance. The opaqueness of their methods make me the more convinced that this approach would require validation in a prospective clinical setting.

We agree with the reviewer that clear indication of 'success' should have been indicated for the performance in our validation cohort. Statistically speaking, validation of an algorithm means achieving statistically significant performance in un-seen datasets. There have been several improvements in the revised version of our work including:

- (1) The expansion of our datasets governing 3 medical centers and one multi-center consortium. We dedicated data from the external medical center (Dana Farber) specifically as an unseen cohort to benchmark the proposed model.
- (2) We performed 1:1 propensity score matching to mimic randomized control trial during model discovery and validation, which is a well-used statistical technology to bridge the missingness of randomized data.

- (3) We showed several performance metrics that highlight the valuable findings in our external validation cohort such as:
1. The measurement of progression risk reduction, when a subgroup of patients was treated as recommended in comparison those who were treated against recommendation (**Table 3**)
 2. Median survival difference, when patients received chemotherapy as recommended by the system (**4-5 months longer**)
 3. Two-year PFS stratification, using only the prediction of response to therapy within the first 3-months into the intervention (**Fig.3**)

We hope the reviewer will find these additional datasets and analyses would provide readers with a clear and quantitative evaluation about the potential of our model to stratify patients under different treatment arms through rigorous validation.



- f. The figures are not nearly of a standard that would be expected for publication in this journal. Thank you for the advice. We are committed to make sure our work is up to the standard of this reputable journal and hence, we have made significant efforts to ensure the quality of figures.

5.Reviewer #4 (Remarks on code availability): There was no code provided.

Thank you for your feedback regarding the availability of our codes. We appreciate the importance of transparency and reproducibility in scientific research. We have included the codes in this

revised manuscript to ensure that it is available for review. The codes are now accessible and can be reviewed alongside the manuscript. Additionally, we have made the codes publicly available in a dedicated repository on [<https://github.com/WuLabMDA/A-STEP>], with detailed documentation to facilitate its use and replication of our results. Thank you once again for your valuable feedback.

Response to the reviewers

We thank the reviewers for their time and their thoughtful recommendations to further improve our manuscript. We have now edited the manuscript accordingly and we are more than happy to make any further revisions recommended by the reviewers.

Reviewer #1

(Remarks to the Author):

The authors have addressed many of my comments. The additional external validation data is a nice plus. While I appreciate that the authors cited many subgroup identification methods in the introduction section, I found quite a few of them are only remotely related. It might be a distraction and can actually invite more questions. For example, the references 20-27 are more about biomarker research. Although related, this is not the focus of this paper.

Response: Thank you for your careful reading. We agree that there are indeed references that are more related to biomarker research than subgroup identification. We have removed a significant number of references that are not within the main the focus of this manuscript. In particular, those previously cited from 20-27 are now removed and have been replaced with more relevant citations (references 19-22).

Reviewer #2

(Remarks to the Author):

In this revision, the authors have substantially extended their prior work. Notably, two further large cohorts were added to this study, Dana-Farber and SU2C, and three new prediction methods were also added. The explanation of the methodology has also been improved a lot compared to the previous version. Overall, the authors invested a significant amount of work in this revision, aiming to address as many comments as they could. Unfortunately, another major issue has now been introduced in this revised version, and another major issue has not really been solved. All five methods that were proposed in the original version of this manuscript failed to work in the two validation cohorts. I am afraid that this was the reason why the three other new methods (H-STEP, S-STEP, and A-STEP) were also added, although that is not made clear.

Response: Thank you for your valuable feedback. We agree with the reviewer that the predictive power was lower for external validation. This phenomenon is a common challenge in machine learning, particularly when moving from exploratory analyses to rigorous validation. To address the uncertainties of individual models, we introduced the ensemble learning to increase the confidence of model prediction. We have shown in our previous study how ensemble of different models successfully outperformed the individual models. After meticulous validation, the same observation made in this study in which the ensemble model rectified the challenge of making individual models worked on validation datasets.

To recall, during the first revision, we refined our experimental design to rigorously validate our findings. First, cross-validation was used to improve the robustness of model hyperparameter tuning. Second, these models were replicated in independent dataset, strengthening the confidence in their utility and generalizability. Practically, the ensemble models should not be considered as new models as it was more like a fusion of the results of the existing individual models reported in the original version (first version).

To avoid confusion to the readers, we have now included additional explanation in several places including the overall design, methods and results section to clarify the rationales:

Results

“....The sub-scoring models were fused to generate an ensemble model through different fusion techniques to mitigate the uncertainty inevitable to one type of model”.

“.....Among three different ways to ensemble the individual scoring model, the attention-based method (termed A-STEP: Attention-based Scoring for Treatment Effect Prediction) achieved the best performance with weighted risk reduction ranging from -13% to -23% in the discovery cohort, and -16% to -19% in the external validation cohort (**Table 3**). *By contrast, the individual sub-models, excluding sub-model B, or the other two ensemble models, were not able to achieve a significant level of performance, particularly during external validation.*”

Methods

“...After verifying the stand-alone scoring’s predictive performance, we integrated them into different integrated scoring systems (H-STEP, S-STEP, and A-STEP, **Suppl Methods S4**) to mitigate the potential biases of any single model by leveraging the diversity of different approaches to capture wider patterns in the data.”

“...Using the benefit scores from five distinct scoring systems, we further built an ensemble model using three fusion mechanisms. The impetus was to capture common and potential complimentary signals presented across different stand-alone systems while reducing the individual system’s bias and variance by aggregating across multiple model predictions.”

Discussion

Finally, we also discussed the limitation of machine learning studies and future perspectives in the discussion as the following.

“.....Third, the uncertainty of machine learning model performance arising either from data noise or the model variability itself. In this work, we catered such an uncertainty through ensemble methods by not relying on a single model prediction but rather weight the predictions of multiple models to make final decision as well cross-validation. Future work for uncertainty quantification through techniques such as Monte Carlo to estimate the range of possible outcome might be considered.”

Of these three new methods, only A-STEP can be "validated" in the two new cohorts; however, the p-values for the two arms are 0.016 and 0.0076. Since the authors have now tried 8 methods in total on this data, some kind of p-value correction should be applied to correct for this fact; in that case, these p-values may not remain significant.

Response: Thank you for your suggestion. We have calculated false discovery rate (FDR) using Benjamin-Hochberg method, with cutoff of 0.1, and appended are the results for A-STEP (prognostic p-value and p-interaction) in both discovery and independent cohort which have also added in the abstract:

Table 1: Unadjusted prognostic p-values

Dataset	ICI-Mono (p-value)	ICI-Chemo (p-value)
Discovery	0.048	0.00053
Independent	0.016	0.0076

Table 2: **False discovery rate (FDR)** for Table 1 above.

Dataset	ICI-Mono	ICI-Chemo
Discovery	0.16	0.0034
Independent	0.076	0.034

Table 3: P-interaction

Dataset	Unadjusted p-interaction	FDR
Discovery	2e-06	1.6 e-05
Independent	0.009474	0.076

As is shown in Table 2-3, the relevant p-values largely remain significant even after correction for multiple-hypothesis testing.

So, in total, it could be questionable that the new proposed A-STEP method -- which seems to have been developed using the validation data more than once -- would actually generalize.

Response: Thank you for raising this concern. We want to clarify that there is no information leakage; all single as well as ensemble models were built only on the discovery dataset to fine-tune hyperparameters. Once the models were locked, they were tested on Dana-Farber cohort which was our unseen data.

The methods are tested by determining the survival in the cohorts of those patients who receive the treatment recommended by the model to those who were not treated as recommended by the model. As remarked in the previous manuscript, direct validation of this kind of prediction would require analysis of a _randomized_ rather than retrospective / observational trial. While the authors do employ propensity score matching in an attempt to reduce confounding, we still need to buy that there is no residual confounding, which can be questionable since some important clinical variables seem to be missing (see comment by R4). Unfortunately, this point was not really addressed by the revision. There now is some limited discussion of this issue, but still the claims are presented in stronger language than the results really allow (e.g. a reader would be forgiven to assume that a "proof-of-concept study" of "personalization of immunotherapy" would involve some actual personalization).

Response: Thank you for this critical comment. We agree that true validation of our selection model would require a prospective randomized trial. However, as such prospective data is not available and not feasibly obtainable, our aim is to develop analytic strategies to help leverage the real-world data that gets generated through clinical practice. We further agree that there are significant caveats to real-world datasets, including inherited heterogeneity and missing data issues. We attempted to mitigate against these limitations through propensity score matching (as pointed out), and by focusing on the clinical variables that may impact clinical outcome and were available in all datasets. Again, we agree that important biology may be missed through this approach, but such variables can be incorporated into later model iterations as the technologies that capture these features (e.g. WES, RNA-seq, and single-cell) become more widely deployed.

More broadly, we completely agree with the reviewer that this is a significant limitation of our study, and we appreciate the feedback that the language in our revised manuscript still does not fully convey this fact. We have therefore further softened our language to emphasize these important limitations, and we have removed specific strong terms such as "personalization", "personalized treatment".

“... retrospective analysis can contain unmeasured biases and confounding, and consequently, the models and findings require further assessment using large prospective cohorts. ... we focused on routinely collected clinicogenomic variables, mainly because the use of less common information limits applicability to centers where this information is not routinely collected. Owing to variability in available clinical and genomic data across different datasets, this study utilized a limited number of overlapping features. More multimodal data, including broader mutation profiling, transcriptomic or immune features, and radiomics, may capture biology important to ICI response and can be integrated into future model iterations with likely enhanced model performance...”

Also, we have revised the title to “Machine-Learning Driven Strategies for Adapting Immunotherapy in Metastatic NSCLC: A Proof-of-Concept Study Across Multi-Center Cohorts”

Reviewer #3

(Remarks to the Author):

The authors have significantly revised their original submission based on the feedback from the referees. As a result this revision is much improved and in effect is a brand new paper. I have 3 comments.

(1) Please remove any reference to "artificial intelligence" in the title, abstract and in the paper. You are using machine learning which you note repeatedly throughout the paper. The need to refer to AI is unnecessary. I would strongly encourage you to delete the reference.

Response: Thank you for this suggestion. We have edited the manuscript throughout.

(2) The methods being used are quite sophisticated and details are provided in the supplementary material. I leave it to the Editors to decide if this is an appropriate article for the journal.

Response: Thank you for this comment. To further facilitate the reproduction of the experiments and results, we have updated and restructured the scripts in our Github (). Readers may now use these scripts to replicate the methods in the manuscript step-by-step. We have also provided a tutorial on how one can run the scripts with the intention to help any readers with basic R and statistics knowledge to be able to reproduce the experiments.

(3) Referee 4 made an important observation regarding the original tone and scope of the work. In response the authors wrote:

"Our study is not intended to solve the basic science/ drug development challenges of precision treatment directly, as you correctly pointed out. Our research serves as a proof-of-principle demonstration of how machine learning approaches can be leveraged to address these complex clinical decisions as they arise, particularly in contexts where treatment efficacy cannot be linked to a simple e.g. genomic biomarker . . .

Specifically, we used the ICI-Mono vs ICI-Chemo question as an illustrative example to showcase the potential utility of our approach in guiding treatment decisions, particularly in situations where clinicians face a lack of head-to-head trial data.

We understand and agree that our initial manuscript may have overstated the immediate clinical applicability of our findings. To address this, we have revised the manuscript to clarify that our aim is to complement ongoing drug discovery efforts by providing a statistical and machine learning framework that can help prioritize and evaluate potential treatment options based on patient-specific data. We recognize that our work is one piece of a larger puzzle and does not negate the critical role of basic science, drug development, and clinical trials."

I agree with these comments completely and would like the authors to convey this sentiment at the outset of the paper in the abstract and in the Introduction.

Response: Thank you for highlighting these important points. We have taken your suggestions to embedding these key points in Abstract and Introduction.

Abstract:

“...While this approach does not supplant the need for prospective data, this study serves as a proof-of-principle demonstration of how machine learning approaches can be leveraged to guide clinical decision-making in the context of biological complexity and incomplete data, and to complement new drug development and clinical trials to enhance patient outcomes in the era of immunotherapy.

.”

Introduction:

“... While more research is needed to define the biology underlying ICI sensitivity, one possible reason for the absence of robust biomarkers is the biologic complexity of the tumor-immune axis and potentially non-linear relationships between treatment response and tumor intrinsic and immune features. This biological complexity suggests the need for more sophisticated models that can incorporate and weight multiple variables....”

... “Together, they demonstrate the applicability of machine learning frameworks to the task of prioritizing and evaluating potential treatment options based on patient-specific data, which can complement the larger research enterprise needed to advance clinical care that includes basic science, drug development, and clinical trials”.....

... “While these efforts do not supplant the need for prospective clinical trials or mechanistic preclinical work, our objective is to complement these efforts by providing a statistical and machine learning framework that can help prioritize and evaluate potential treatment options based on existing patient-specific data. Given the dearth of clinical trial data comparing these regimens head-to-head on this topic, we used large multi-institutional datasets...”

...”Collectively, this work demonstrates the application of machine-learning models to inform and guide treatment selection, while also generating preliminary criteria that may help guide selection between ICI regimens for further validation and expansion....”

Response to the reviewers

We thank the reviewers for their time to review our manuscript and provide further suggestions to further improve it. We have now edited the manuscript accordingly and we are more than happy to make any further revisions recommended by the reviewers.

The most important issue I had raised in the previous iteration was the fact that 3 new methods were introduced in that version, as the 5 original methods presented in the initial version failed to work on the validation dataset. While the authors seem to agree with this assessment, they respond that the three new methods should in fact not be considered "new methods" since they are ensembles of the earlier ones. I disagree with this statement because in ensemble learning one introduces additional parameters that can be fitted, meaning that the capacity of the models increases and they can generate different results.

The new version of the paper now presents this approach as if it had been the plan all along and does not clearly state that these models were only designed after the initial ones failed to work (they were not in the original version of the manuscript).

Response: Thank you for pointing this out. We agree with the reviewer and have clarified this point in the Methods/ Discussion section.

Method

..." When the stand-alone scoring's predictive performance did not achieved the desired performance during validation, we have introduced different strategies to integrate them (H-STEP, S-STEP, and A-STEP, Suppl Methods S4)"...

Discussion

..." Third, the performance of a machine learning model can fluctuate across different datasets, often due to population shifts or limitations in its generalizability. This was particularly evident when we introduced an external validation dataset during the revision process. To address this limitation, we employed ensemble learning, which combines different models to reduce variance and uncertainty"...

In addition, the authors have performed a p-value correction. I am unable to verify this calculation as I cannot find the individual p-values reported anymore, nor information on which models were included in this correction (just the final 3 or all 8?). However clearly the corrected p-values show that the fact that only 1 out of the 3 ensemble models reaches significance (in fact only 1 out of the 8 total tested ones) does not remain significant once the amount of tested models is taken into account. The authors choose a relatively lenient correction method (FDR instead of

Bonferroni) and an unusual "significance" (in fact FDR) threshold of 0.1 to state significance anyway.

In total, the current manuscript version still presents a somewhat overly positive view of the authors' findings, and the evidence presented that this approach actually works is not particularly strong from a statistical perspective.

Response: Our team, led by biostatistician Dr Jack Lee, has implemented the strict Bonferroni correction per your suggestion. The decision to set the FDR cutoff at 0.1 reflects the exploratory nature of this study, which aims to identify potential findings for future validation. We have acknowledged this in the Discussion section. To enhance transparency and facilitate further validation, we have publicly shared both the data and models.

Suppl Table S1 presents the results of the log-rank test in the external validation cohort for various models. In the ICI-chemo arm, the Bonferroni-adjusted p-value for A-STEP is 0.061, which falls below the pre-specified 10% FDR threshold and is thus considered statistically significant. However, the corresponding adjusted p-values do not reach significance at the 0.05 level. In the ICI-mono arm, A-STEP has the smallest p-value, indicating a trend toward statistical significance.

Additionally, we have conducted a new experiment to directly assess the predictive performance of model stratification, by calculating the interaction p-value and applying Bonferroni correction.

Supp Table S2 displays the interaction p-values between models and treatment arms during external validation. For A-STEP, the Bonferroni-corrected p-value is 0.076, which meets the 10% FDR criterion and is considered statistically significant, though it does not reach significance at the 0.05 level.

Suppl Table S1: Log-rank p-values for two treatment arms during external validation

Models	ICI-mono			ICI-chemo		
	Unadjusted	BH	Bonferroni	Unadjusted	BH	Bonferroni
A	0.970	0.980	1.000	0.920	0.920	1.000
B	0.750	0.980	1.000	0.770	0.920	1.000
C	0.970	0.980	1.000	0.920	0.920	1.000
D	0.980	0.980	1.000	0.910	0.920	1.000
E	0.019	0.076	0.152	0.008	0.034	0.067
H-STEP	0.880	0.980	1.000	0.480	0.920	1.000
S-STEP	0.630	0.980	1.000	0.400	0.920	1.000
A-STEP	0.016	0.076	0.128	0.008	0.034	0.061

Suppl Table S2: Interaction p-value during external validation

Models	P-interaction		
	Unadjusted	BH	Bonfer-roni
A	0.813	0.813	1.000
B	0.293	0.781	1.000
C	0.813	0.813	1.000
D	0.453	0.813	1.000
E	0.083	0.335	0.670
H-STEP	0.587	0.813	1.000
S-STEP	0.672	0.813	1.000
A-STEP	0.009	0.076	0.076

Discussion

...."Fourth, we set FDR threshold at 0.1 (10%) to reflect the exploratory nature of this study, but we also implemented a strict Bonferroni correction to asses statistical significance level.".....